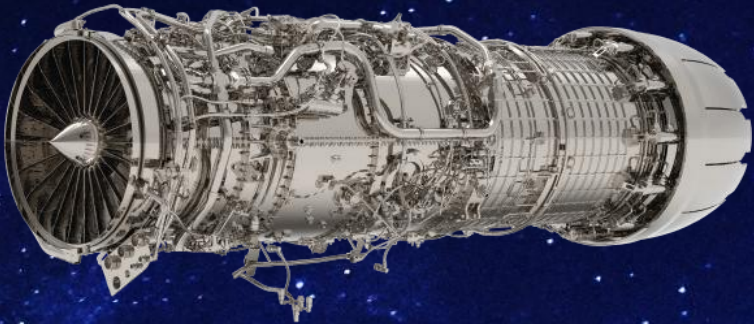


TEKNOFEST 2026

Gas Turbine Engine Variable Exhaust Nozzle

Yasemin Sarıoğlu
Ahmet Onur Şahin



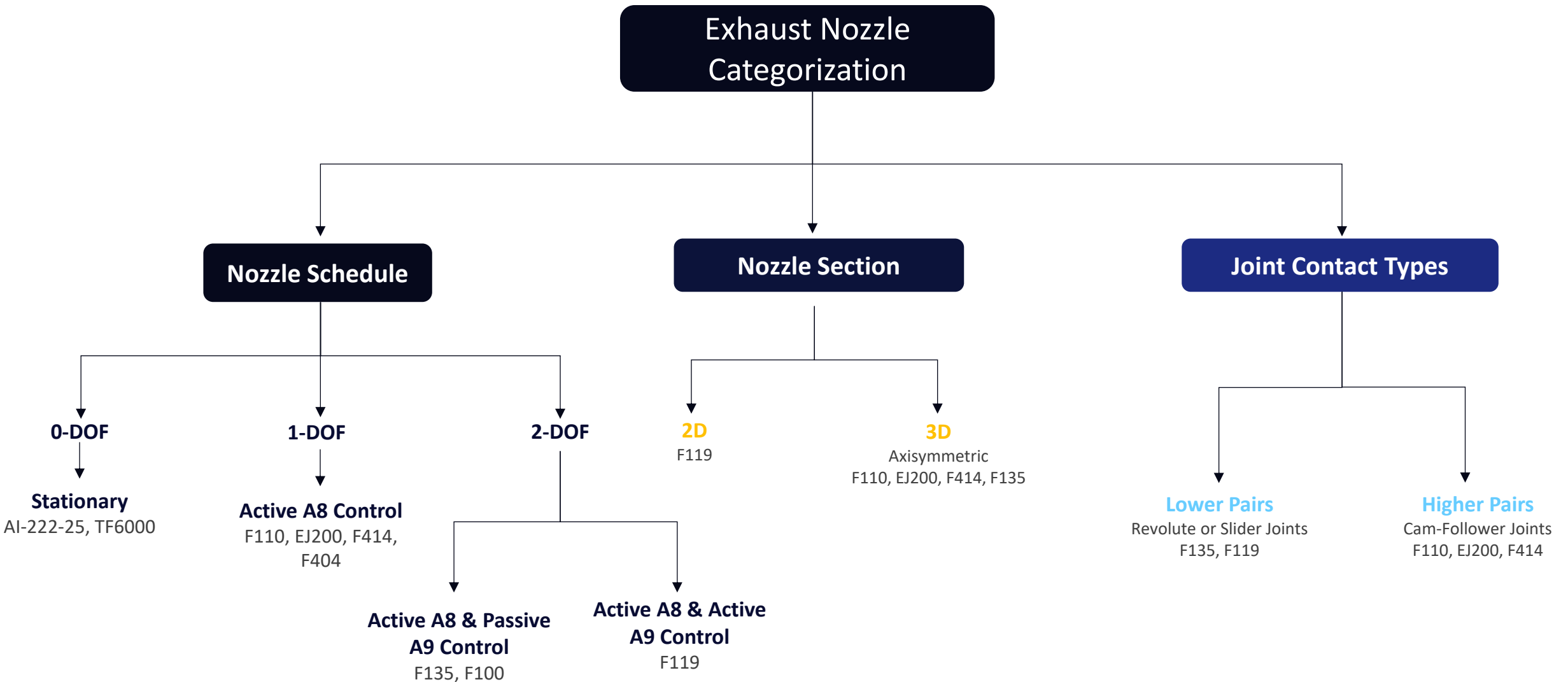
- Variable Exhaust Nozzle Functions
- Nozzle Types
- Main Components
- Mechanism Design
 - Kinematic Synthesis Procedure
 - Part Design
 - Joint Design
- Design Criteria
 - Strength
 - Life
 - Buckling

What is the Variable Exhaust Nozzle?

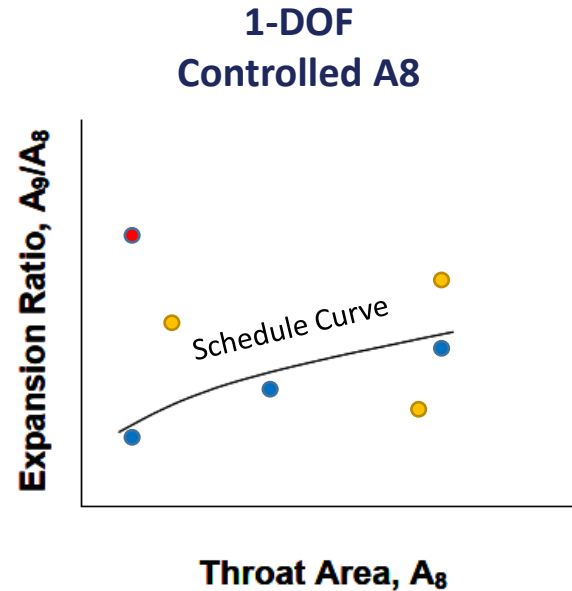


The tailpipe regulates the speed of the exhaust gas, thus providing the necessary thrust. **In military engines with afterburners, the tailpipe is adjustable/variable and is opened during AB activation to accommodate the large volume of hot gas.**

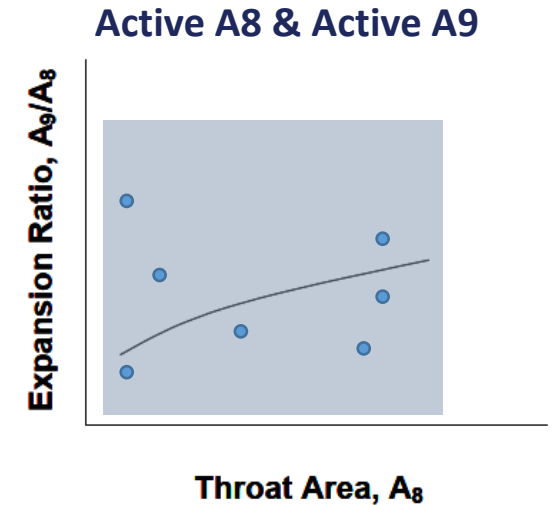
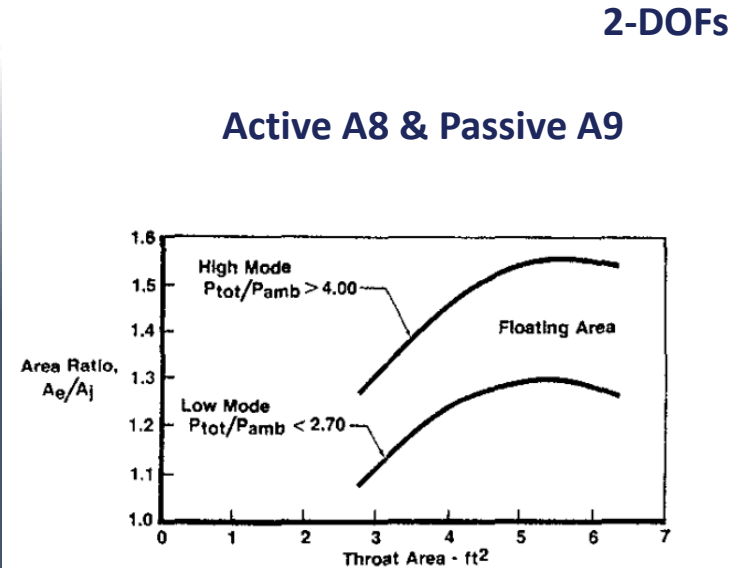




Nozzle Schedule (1-DOF & 2-DOF)



- A8 is actively controlled by one actuating system.
- A9 is geometrically tied to A8, not controlled. (a.k.a Fixed Schedule)
- Schedule profiles are similar, depends on link lengths and/or positions
- e.g. F110, F414, RM12, EJ200

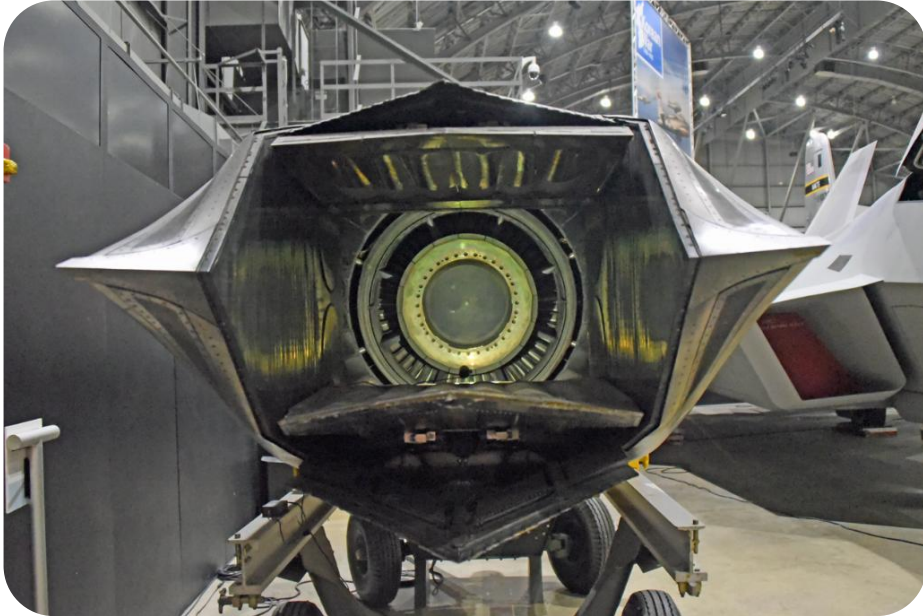


- A8 is actively controlled by one actuating system.
- A9 is controlled whether an active or passive system, so, different A9 values can be provided for the same A8 value.
- F135, F100, F119.

Nozzle Schedule (1-DOF & 2-DOF)

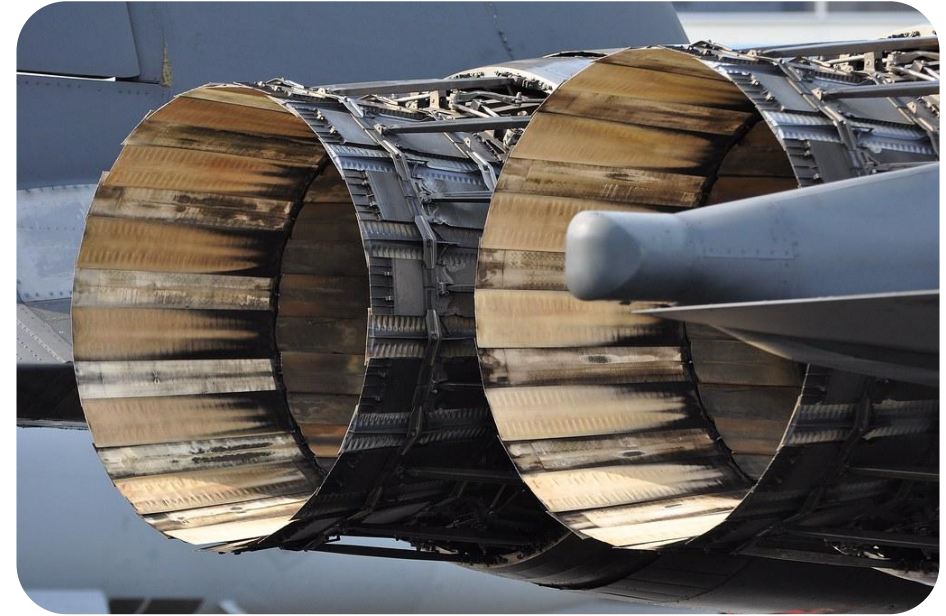


2D Section



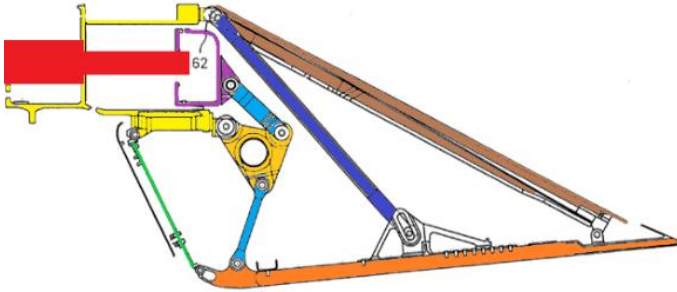
- Can offer thrust vectoring in 1 axis.
- e.g. F119 (F22 Engine)

3D Section



- Axisymmetrical exit section.
- Can offer thrust vectoring in 2 axis.
- Harder to synchronize flaps.
- e.g. F135, F100, F110, AI-31F

Lower Pairs



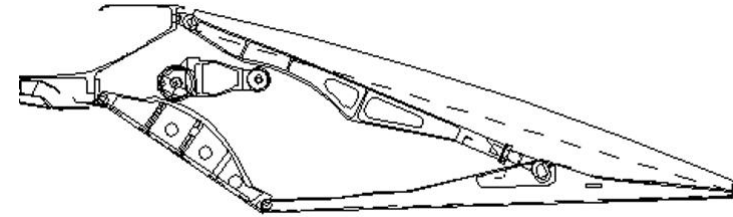
Revolute or Slider Joints

- ✓ Joints are easier to design.
- ✓ Easy to manufacture.
- ✓ Easy to lubricate.
- ✓ Friction effect on the connection points is low.
- ✓ Easier to maintain.

More but smaller parts than other options.

- ❖ Difficult to assembly.
- e.g. F135, AI-31F, F100**

Higher Pairs

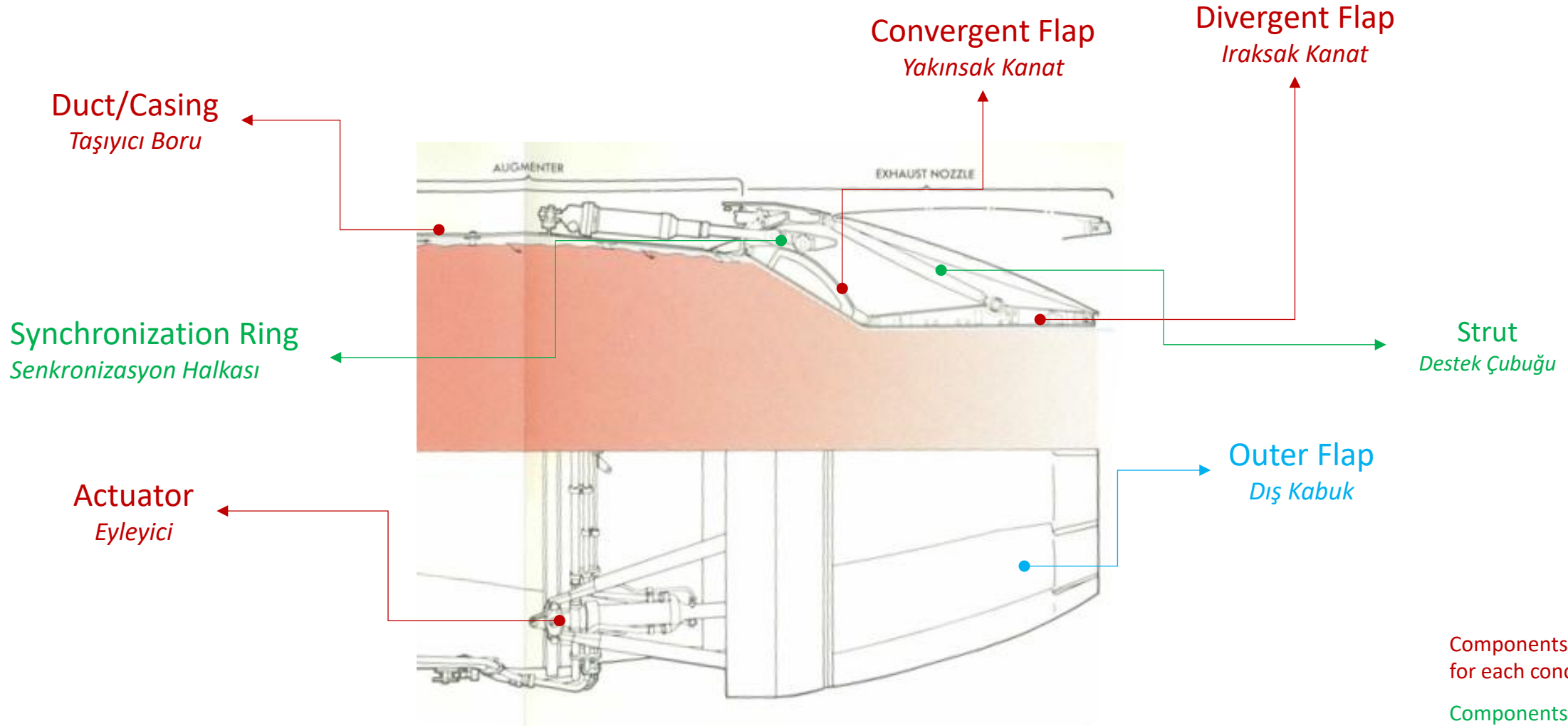


Cam-Roller Joints

- ✓ Mechanism is easier to design and manage.
 - ✓ Less moving parts.
 - ✓ Easier to assembly.
- ❖ Hard to manufacture.
- ❖ Line contact without lubricants.
- ❖ Friction effect on the connection points is high.
- ❖ High contact stresses on contact points.

e.g. F110, EJ200, F414

VEN Components – 2D



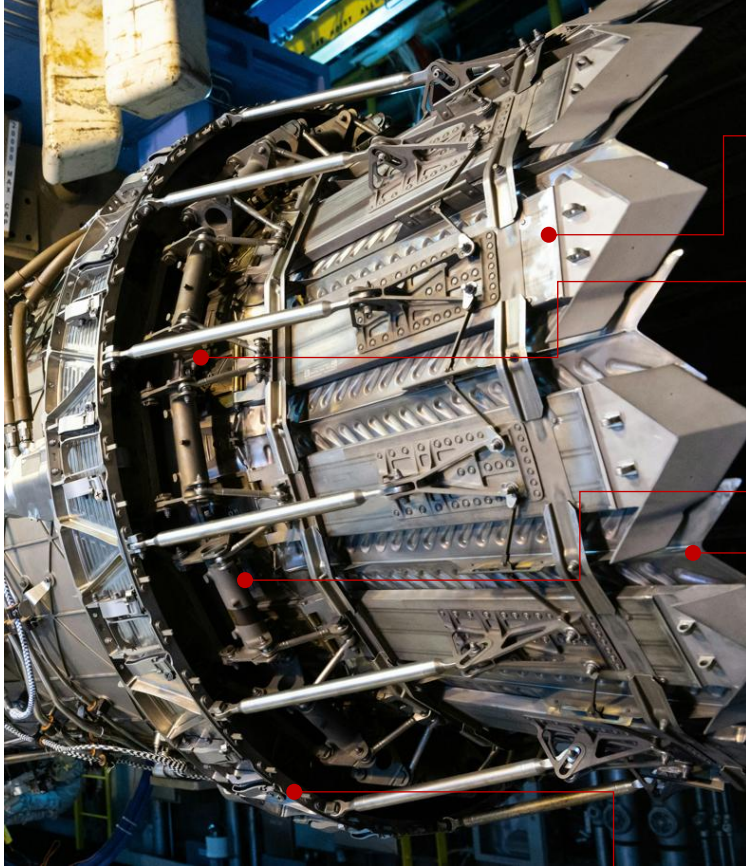
+ Bağlantı Braketleri ve Bağlantı Elemanları (Civata, Somun, Burç, Pul vs.)

Components that are required for each concept

Components optional to the selected concept

Components that are not expected to be designed

VEN Components – 3D



F135 (Görsel Google'dan alınmıştır)

Components that are required
for each concept

Parts required to complete an enclosed flowpath;

Divergent Flap
Iraksak Kanat

Convergent Flap
Yakınsak Kanat

Convergent Seal
Yakınsak Conta

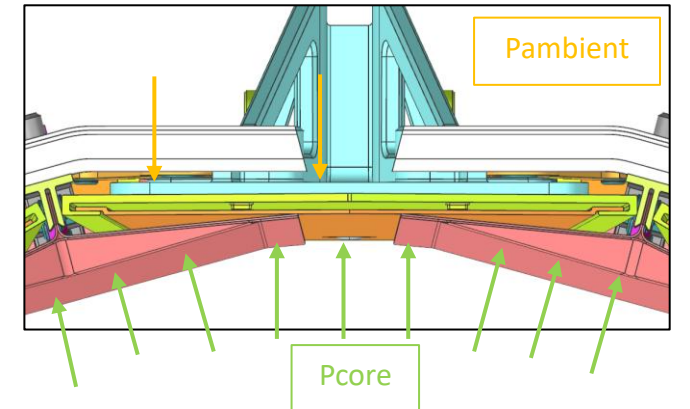
Divergent Seal
Iraksak Conta

Duct/Casing
Taşıyıcı Boru

Master elements: Parts that are directly driven by actuators.

Slave elements : Parts that are positioned according to the master parts movements

P.S. It is important to design interfaces so that each part has enough Degree of Freedom during mechanism motion.

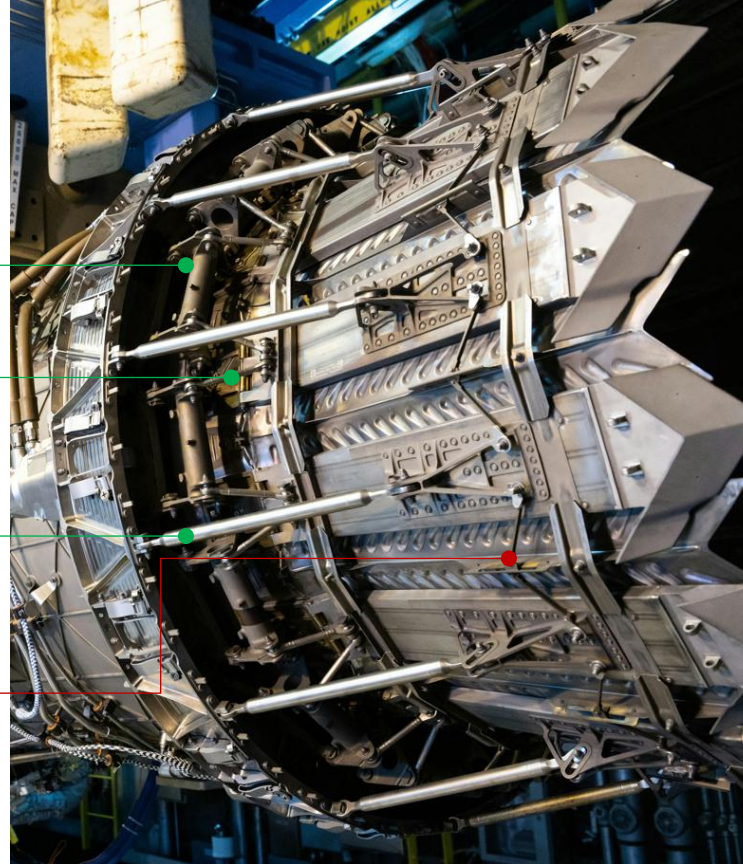


VEN Components – 3D

Mechanism
Elements
Mekanizma Elemanları

Strut
Destek

Synchronization
Elements
*Çevresel Senkronizasyon
Elemanları*



F135 (Görsel Google'dan alınmıştır)

Design Recommendation: Flow path parts design and mechanism elements design can be separated.

During flowpath parts design, make sure;

- Each part has enough Degree of Freedom during mechanism motion.
 - Clearances are minimized between parts throughout the movement (min to max A8) to prevent leakage of air.
 - Parts do not interfere with each other at any position.
- ❖ Mechanism elements can later be built on top of these elements.
- ❖ Synchronization elements' design are not mandatory. However, doing so may earn extra points.

+ Bağlantı Braketleri ve Bağlantı Elemanları (Civata, Somun, Burç, Pul vs.)

Components that are required for each concept
Components optional to the selected concept
Components that are not expected to be designed

Mechanism Design

Degree of Freedom : DOF is equal to the number of independent parameters which are needed to uniquely define mechanism's position in space at any instant of time.

- The DOF of VEN mechanism should be determined according to the performance requirements.

Greubler Equation

$$M = 3(L - 1) - 2J_1 - J_2$$

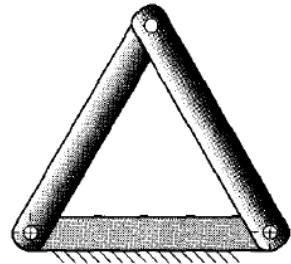
M = degree of freedom or mobility

L = number of links

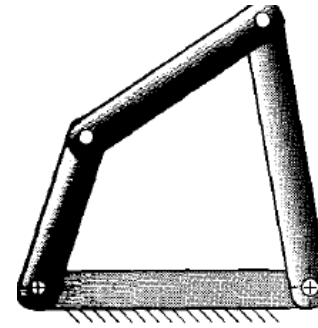
*J*₁ = number of 1DOF (full) joints

*J*₂ = number of 2 DOF (half) joints

For a planar mechanism

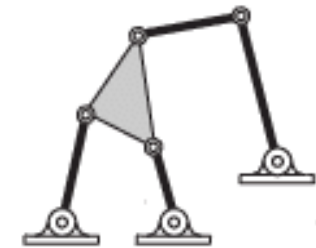


Structure—DOF = 0

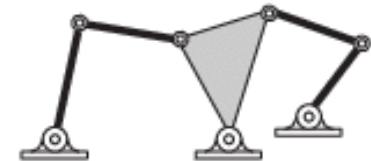


Mechanism—DOF = + 1

**4-bar mechanism
DOF : 1**



Stephenson III



Watt II

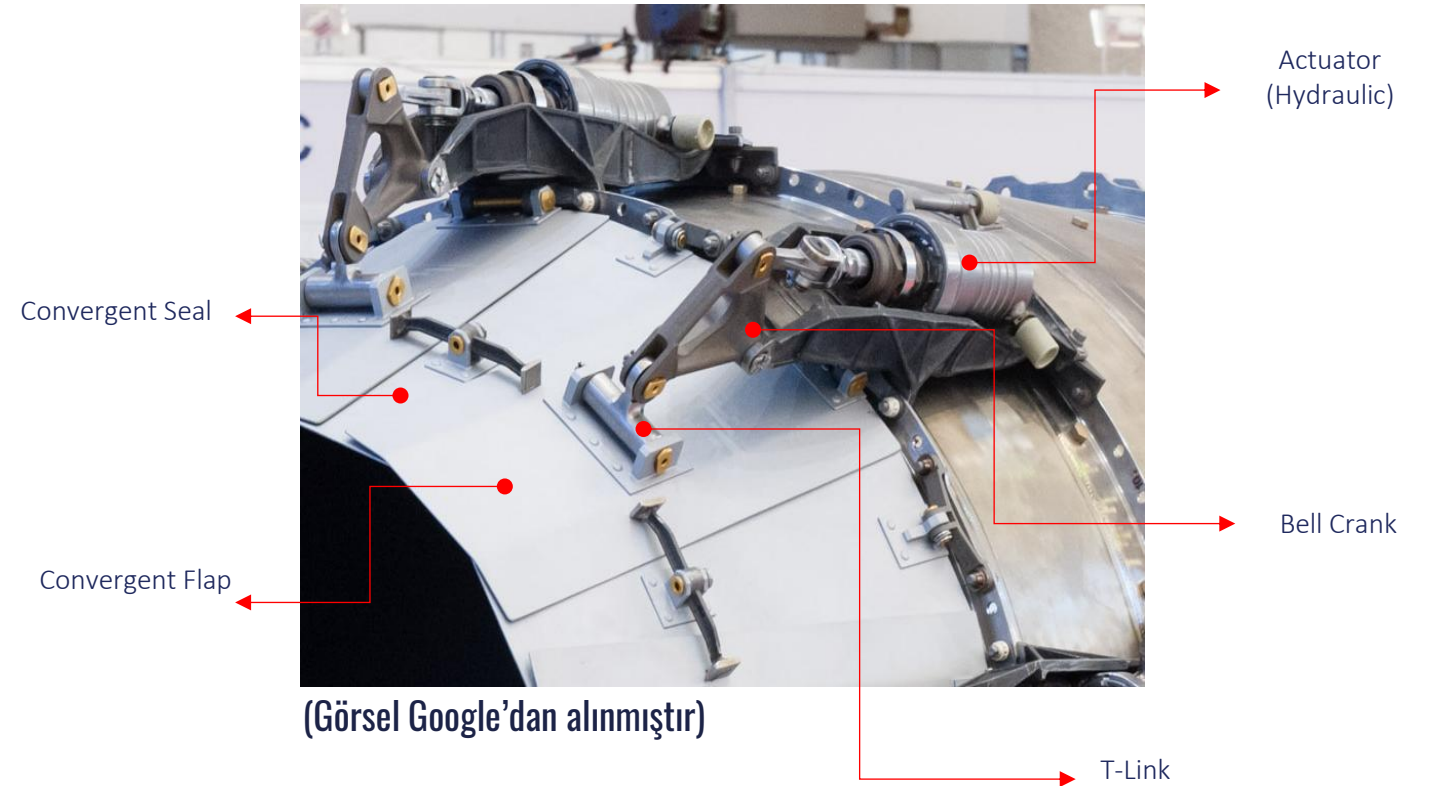
**6-bar mechanisms
DOF : 1**

Mechanism Design

AI-322F Engine Exhaust



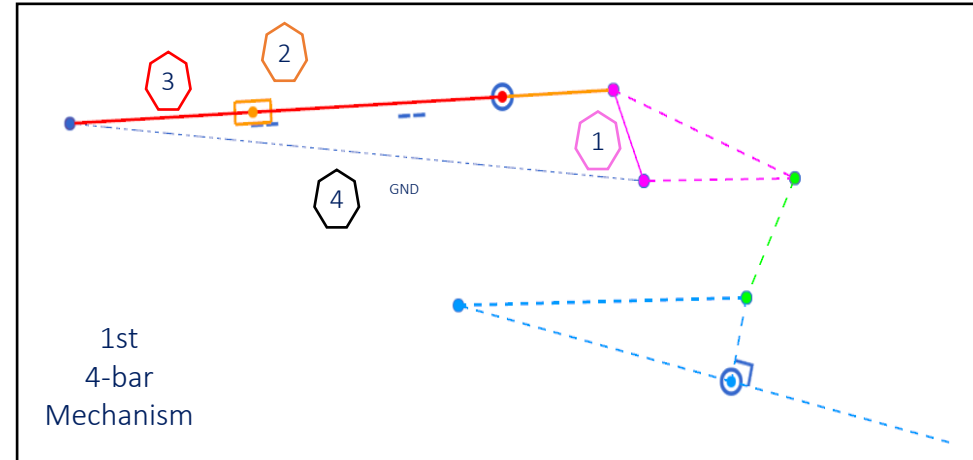
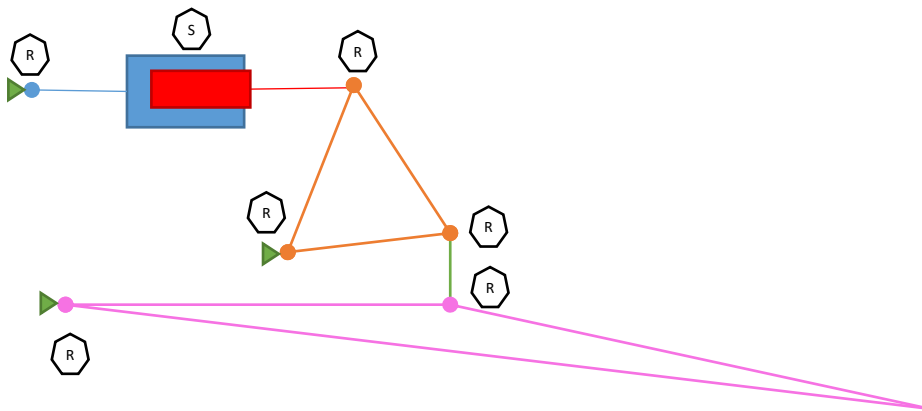
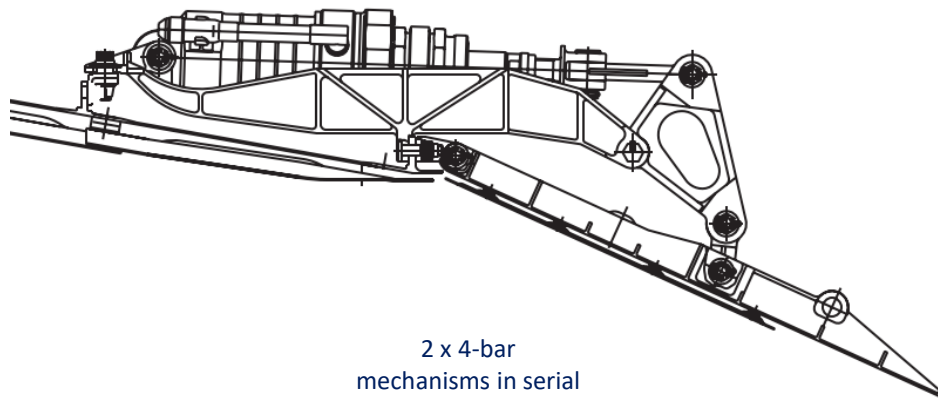
- Convergent Nozzle
- 2D planar mechanism
- One actuator per flap
- 9 flaps – 9 actuators



Mechanism Design – Convergent Section

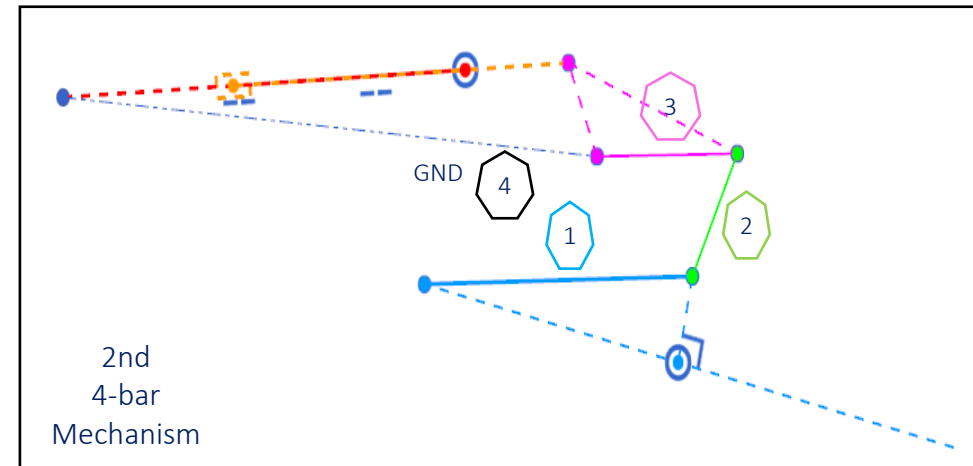
AI-322F Engine Exhaust

(Görsel Google'dan alınmıştır)



$$M = 3(L - 1) - 2J_1 - J_2$$

$L = 4, J_1 = 4, J_2 = 0$
 $M = 1$

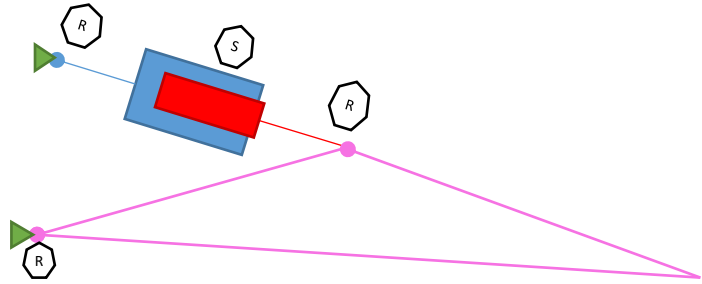


$$M = 3(L - 1) - 2J_1 - J_2$$

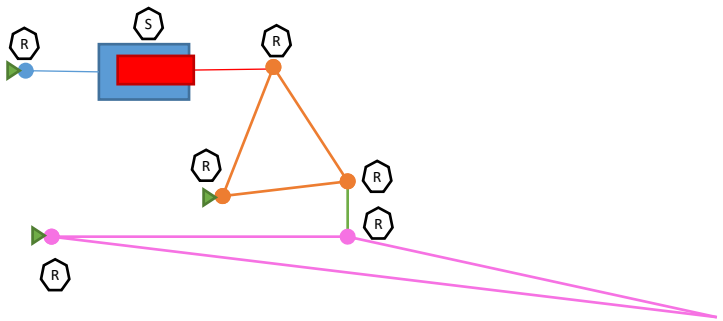
$L = 4, J_1 = 4, J_2 = 0$
 $M = 1$

1 DOF system
Input = Actuator Stroke, Output = A8

Mechanism Concepts – Convergent Section

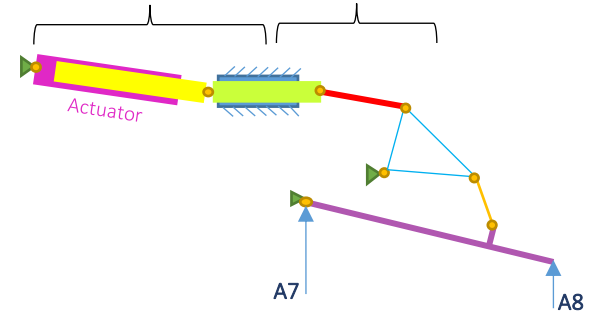


1 x 4-bar mechanism



2 x 4-bar mechanisms in serial

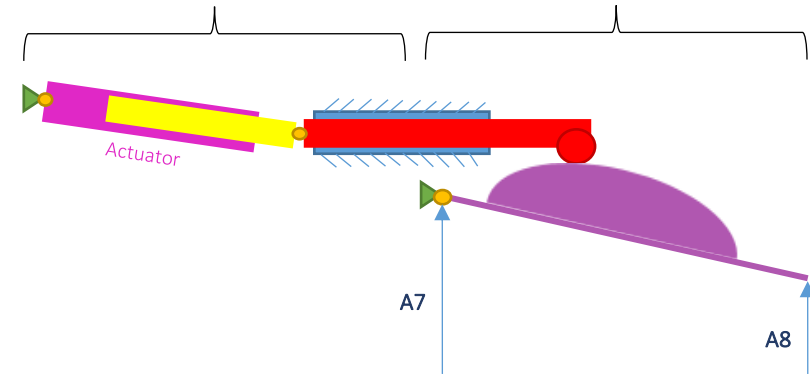
1 . 4-bar Mechanism 2 . 4-bar (Slider-Crank) Mechanism



3 x 4-bar mechanism

1 . 4-bar (Slider – Crank) Mechanism

2 . Cam Follower Mechanism

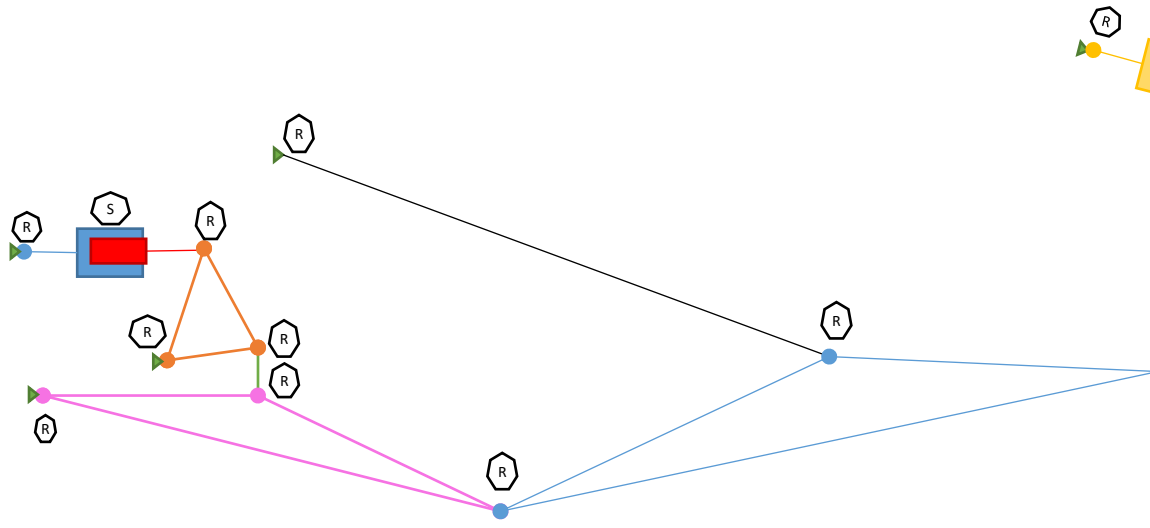


4-bar + Cam-Follower mechanisms in serial

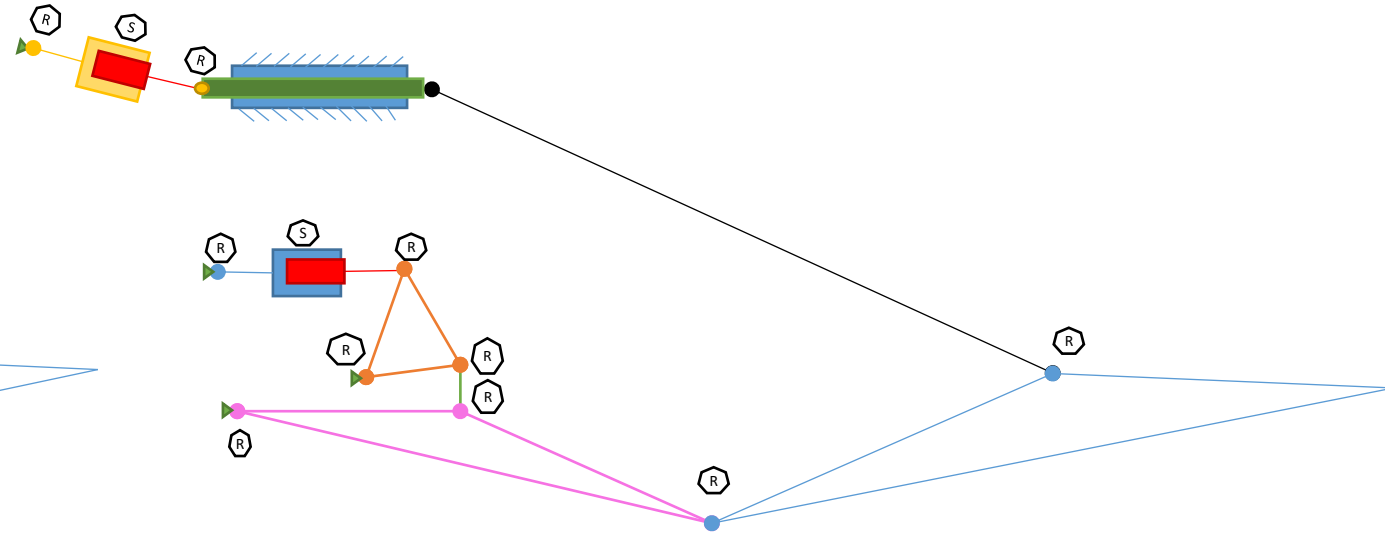
- ▶ Ground
- Revolute Joint

Mechanism Concepts – Divergent Section

- ▶ Ground
- Revolute Joint

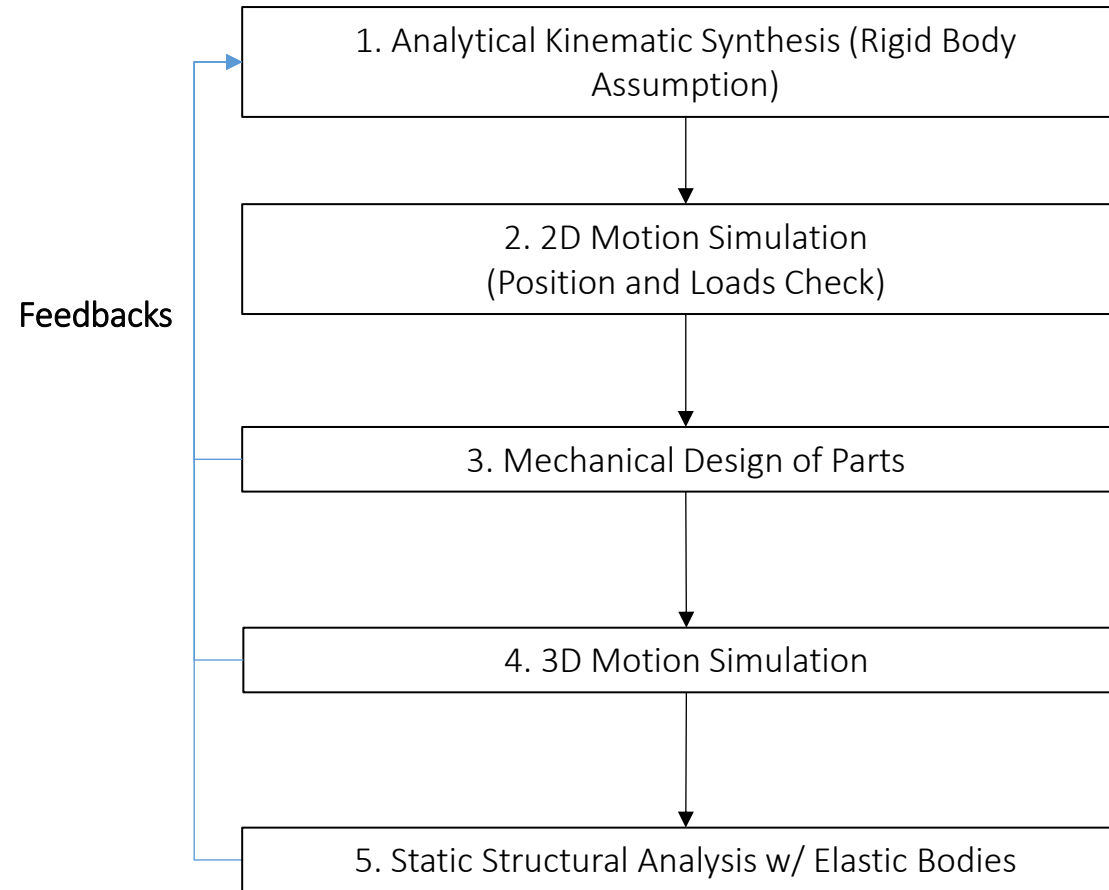


1 DOF System

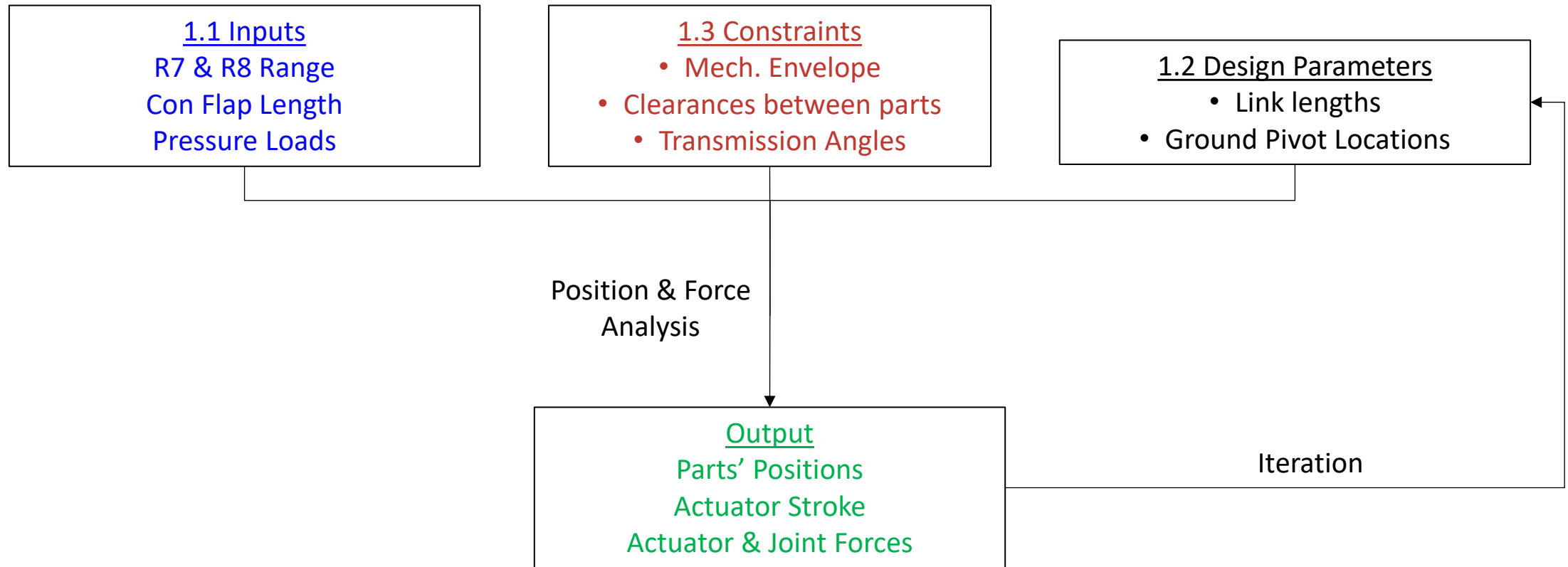


2 DOF System

Mechanical Design Procedure



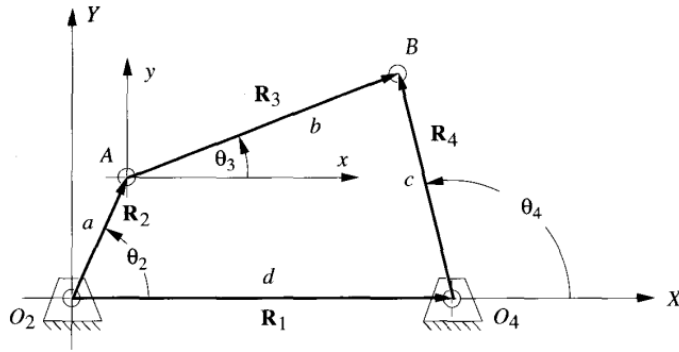
Kinematic Design Procedure



P.S. Velocity and Acceleration analyses are not required because parts have considerably low weight. Dynamic effects can be neglected in such systems.

Kinematic Calculations

Position Analysis



$$A_x = a \cos \theta_2$$

$$A_y = a \sin \theta_2$$

Pisagor's Theorem

$$b^2 = (B_x - A_x)^2 + (B_y - A_y)^2$$

$$c^2 = (B_x - d)^2 + B_y^2$$

Solving Point B Position

$$B_x = \frac{a^2 - b^2 + c^2 - d^2}{2(A_x - d)} - \frac{2A_y B_y}{2(A_x - d)} = S - \frac{2A_y B_y}{2(A_x - d)}$$

$$B_y^2 + \left(S - \frac{A_y B_y}{A_x - d} - d \right)^2 - c^2 = 0$$

$$B_y = \frac{-Q \pm \sqrt{Q^2 - 4PR}}{2P}$$

$$P = \frac{A_y^2}{(A_x - d)^2} + 1$$

$$R = (d - S)^2 - c^2$$

$$Q = \frac{2A_y(d - S)}{A_x - d}$$

$$S = \frac{a^2 - b^2 + c^2 - d^2}{2(A_x - d)}$$

Solving θ_3 & θ_4

$$\theta_3 = \tan^{-1} \left(\frac{B_y - A_y}{B_x - A_x} \right)$$

$$\theta_4 = \tan^{-1} \left(\frac{B_y}{B_x - d} \right)$$

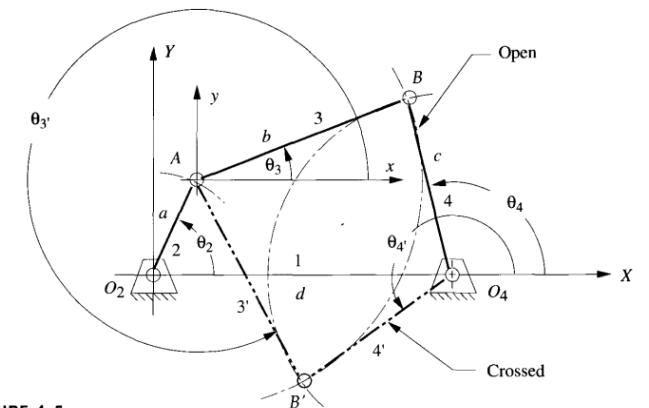


FIGURE 4-5

Graphical position solution to the open and crossed configurations of the fourbar linkage

Source: DESIGN OF MACHINERY AN INTRODUCTION TO THE SYNTHESIS AND ANALYSIS OF MECHANISMS AND MACHINES, Robert L. Norton

Kinematic Calculations

Force & Moment Analysis

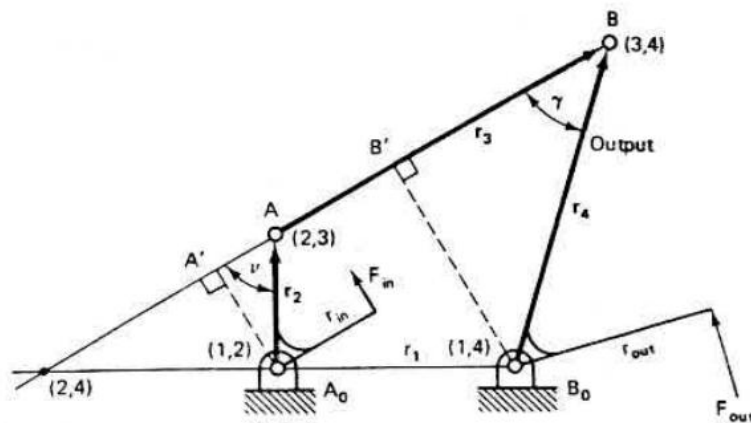


Figure 3.78 Correlation of M.A. with transmission angle γ and crank-coupler angle ν in a four-bar mechanism. See Eq. (3.69).

Mechanical Advantage

- Mechanical Advantage is a measure of mechanism efficiency in load transmission.
- Once positional analysis has been done, mechanical advantage can be analytically calculated via trigonometric relations.

$$\text{M.A.} = \frac{F_{\text{out}}}{F_{\text{in}}}$$

$$m_A = \left(\frac{T_{\text{out}}}{T_{\text{in}}} \right) \left(\frac{r_{\text{in}}}{r_{\text{out}}} \right)$$

$$\text{M.A.} = \frac{r_{\text{in}}}{r_{\text{out}}} \left(\frac{r_4 \sin \gamma}{r_2 \sin \nu} \right)$$

Part Designs

Linkage dimensions are determined by kinematic synthesis.

Parts should be designed according to the loading on parts. A total thickness increase is an ineffective way of mechanical design due to the weight.

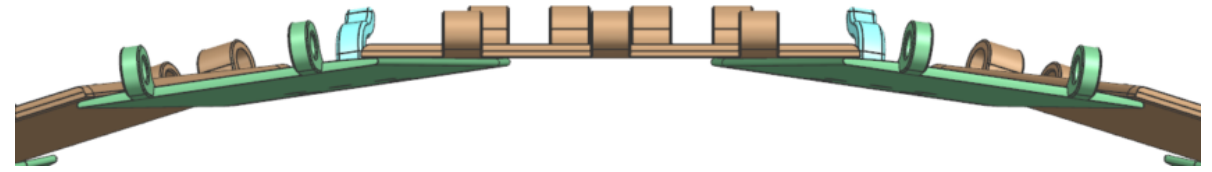
Main loads on VEN mechanism;

- **Pressure Loads**
- Thermal Loads
- Frictional Loads
- Mechanical Loads
- Maneuver Loads

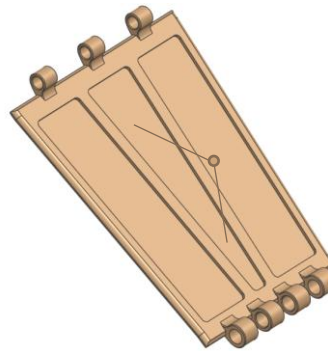
Mechanism parts are working under compression.

Parts that are transmitting torque like a bell crank are exposed to bending stresses.

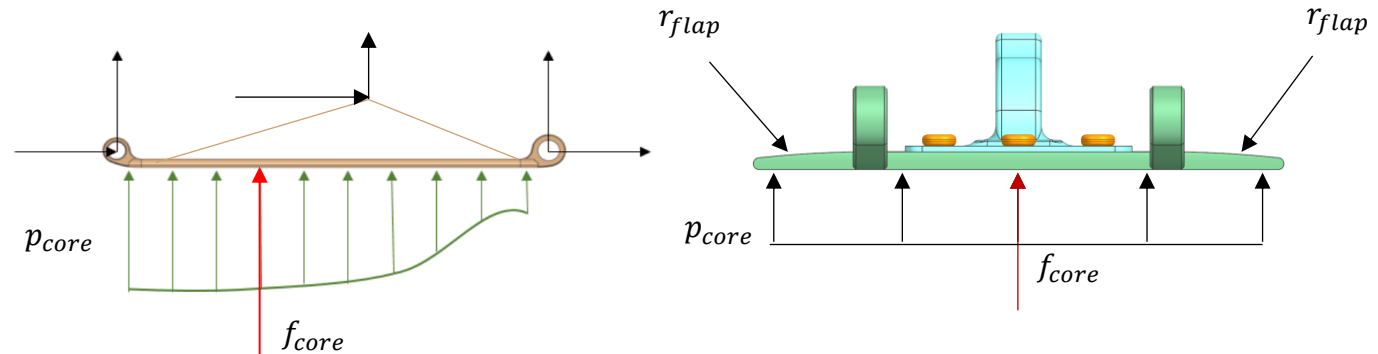
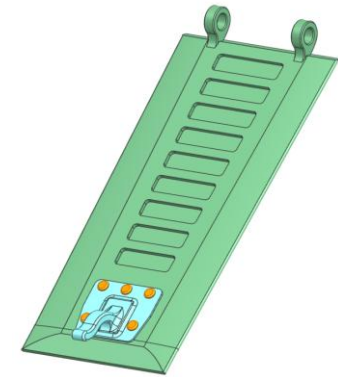
The deformation under loading should be minimized in order to prevent linkage length changes, A8 & A9 changes.



Convergent Flap

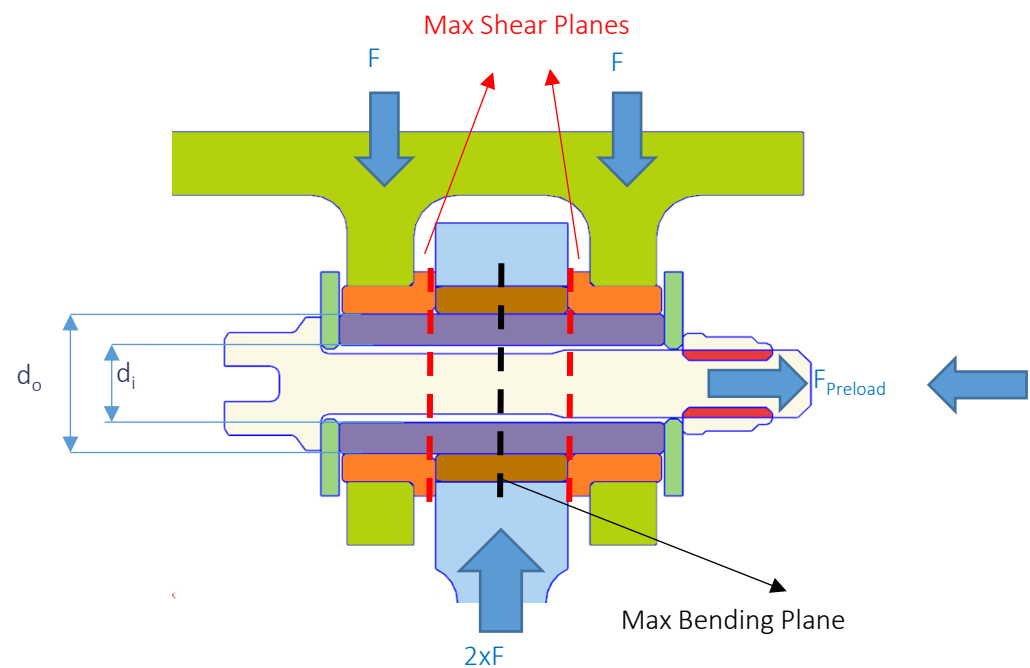


Convergent Seal



Joint Design & Loadings

Revolute Joint Construction Example



P.S. Fit type selection is important in terms of joint freedom and stresses between joint elements. They are expected to be shown in the design report.

Strength for Pin

Bending Requirement

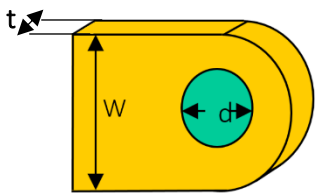
$$RF_{Bending} = \frac{\text{Max Bending Stress}}{\text{Allowable Stress of Pin Material}}$$

Shear Requirement

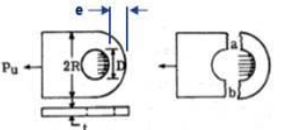
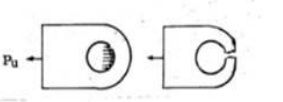
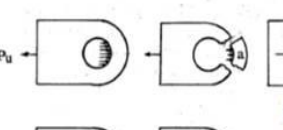
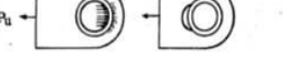
$$RF_{Shear} = \frac{\text{Max/Mean Shear Stress}}{\text{Allowable Stress of Pin Material}}$$

Axial Requirement

$$RF_{Tension} = \frac{\text{Axial Stress with Preload}}{\text{Allowable Stress of Pin Material}}$$



Strength for Lug

Net Section	$\sigma_n = P_u / (2R-D)t$	
Hoop Tension	$\sigma_h = P_u / et$	
Shear Tear Out	$\sigma_s = P_u / 2Lt$	
Bearing	$\sigma_{br} = P_u / Dt$	

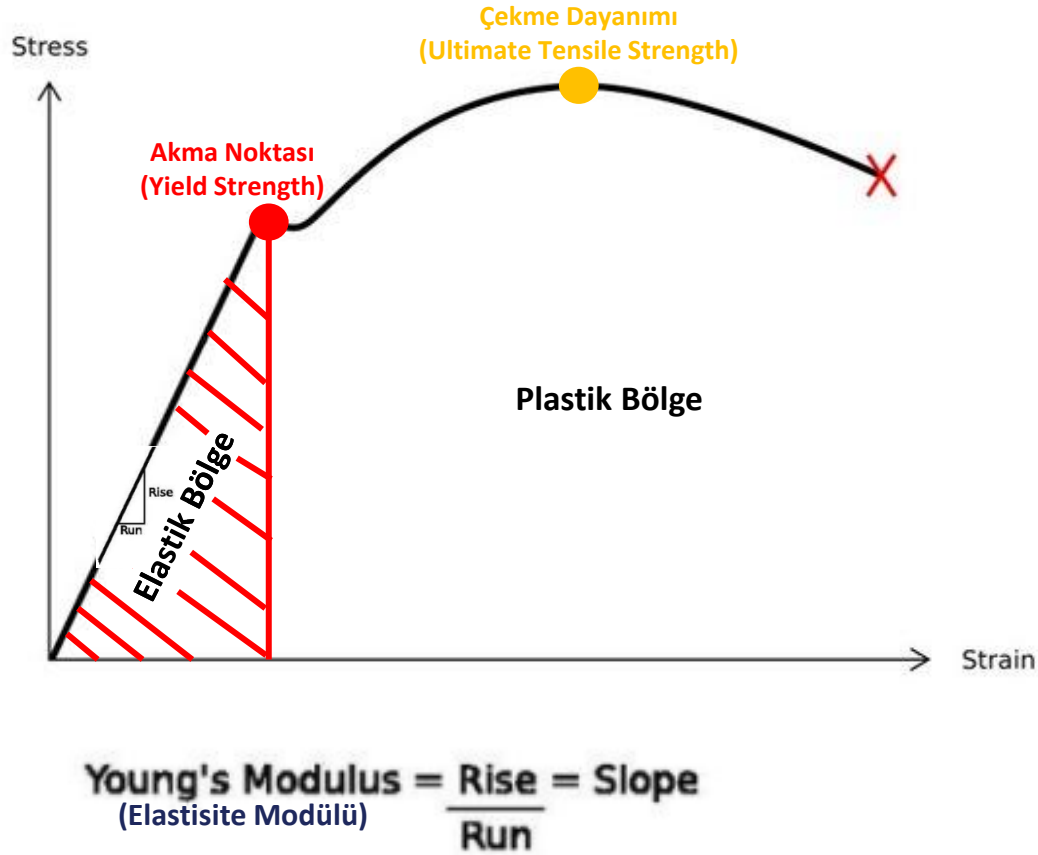
Design Selections

- ❖ Number of flaps
- ❖ Mechanism concept
 - 4-bar or 6-bar or Cam-roller type
 - Planar or spatial
- ❖ Mechanism dimensions
- ❖ Usage of synchronization ring or not
- ❖ Part dimensions
- ❖ Manufacturing methods
- ❖ Joint construction
- ❖ Number of actuators

Mechanical Design Criteria

Statik Parçaların Mekanik Tasarım Kriterleri	
Strength (Dayanım)	Parçaların çalışma anında karşılaşabileceği en kötü durumda bütünlüğünü koruması istenmektedir. Bu sebepten akma sınırını (Yield Strength) aşmaması beklenmektedir. Ayrıca bir hata durumunda (Failure) platformun güvenli bir şekilde inmesi sağlanmalıdır bu nedenle çekme dayanımı (Ultimate Tensile Strength) aşılmamalıdır.
Ömür (LCF-HCF)	Parçaların değişken yükler karşısında bütünlüğünü koruması beklenmektedir. Düşük çevrim yorulma (Low Cycle Fatigue) ve şartnamede belirtilen ister sağlanmalıdır (İster:500 Saat) (1 çevrim=1 saat alınmalıdır.)
Burkulma (Buckling)	İnce cidarlı parçalar ve basma kuvvetlerine ya da yüksek basınç yüklerine maruz kalan parçalar için kritiktir.
Titreşim (Vibration)	Motorun çalışma aralığında bir doğal frekans istenmemektedir. Çünkü bu durum parçalarda değişken gerilmeler yaratıp yüksek çevrim yorulmasında kırıma götürebilir o yüzden bunlardan kaçınmak gerekmektedir.
Çatlak İlerleme Crack Propagation	Tüm parçalar için kritiktir, Hasar toleransı kapsamında herhangi bir parçada çatlak oluşması durumunda parçanın çatlak ilerleme ömrünün ilk bakım zamanına kadar yetmesi istenir.
Sürünme (Creep)	Yüksek sıcaklığa ve sabit bir yüke maruz kalan parçalarda görülen bir durumdur. Kontrolleri sağlanıp ömür isterini sağlaması beklenir.
Üretilebilirlik	Malzemelerin üretim formlarına ve geometrilerine bakarak bir tasarımın çıkartılması beklenmektedir.
Montaj Edilebilirlik	Montajlama sonunda bir serbestlik dereceleri kısıtlanmış veya serbest bırakılan yerler önceden düşünülerek bırakılmış olmalıdır. Aynı zamanda civata gibi bağlantı elemanlarının montajı için de torklama cihazlarının o bölgeye erişebilmesi beklenmektedir.

Strength Criteria



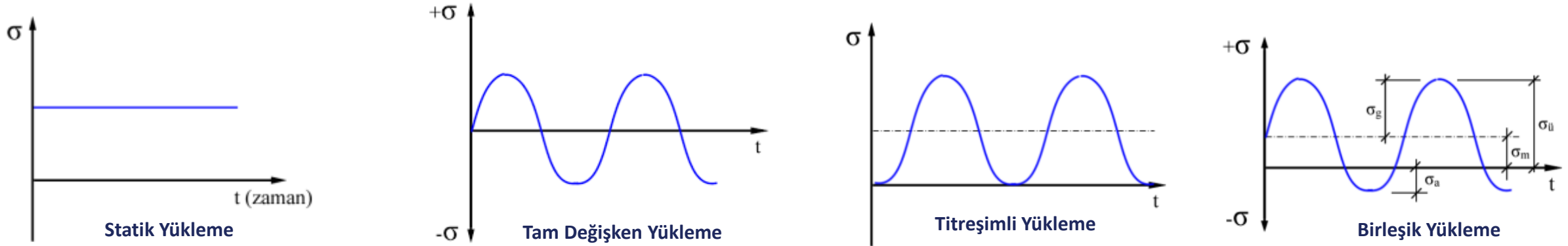
$$\sigma_{Mises} = \sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2 - \sigma_x\sigma_y - \sigma_y\sigma_z - \sigma_z\sigma_x + 3\tau_{xy}^2 + 3\tau_{yz}^2 + 3\tau_{zx}^2}$$

Normal Çalışma Yüklemeleri için:
 σ Von-Mises Gerilmesi < σ Akma Gerilmesi

En kötü durum için (Normal Çalışma Yüklemeleri x 1.5):
 σ Von-Mises Gerilmesi < σ Çekme Gerilmesi

Life - Fatigue

Bir elemana etkiyen kuvvetin değeri kısa zaman aralığında değışiyorsa dinamik yük söz konusudur ve eleman bir süre sonra yorulma sonucu kırılır. Bu nedenle dinamik zorlanmalarda, eleman yalnız şekil değıştirme ve kırılmaya göre değil, yorulmaya göre de değeriendirilmelidir.



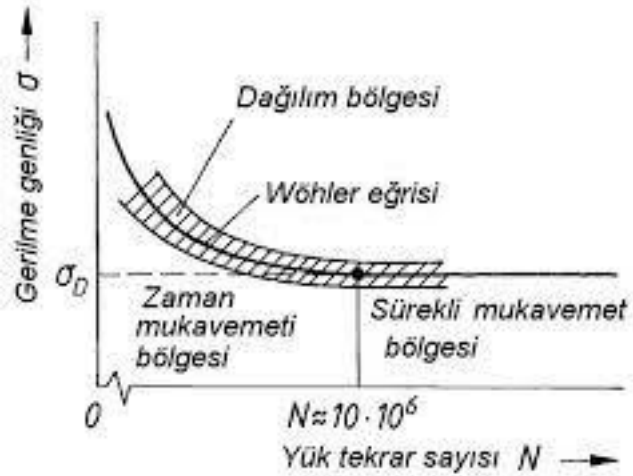
Gerilme değışimi aynı zamanda yüklemenin şeklini de karakterize eder. Temel yükleme şekilleri **Statik**, **Titreşimli** ve **Tam değışken** yükleme şeklinde sınıflandırılmıştır. Bu yükler altında oluşan gerilmeler $\sigma_{üst}$, σ_{alt} , $\sigma_{ortalama}$, σ_{genlik} gerilmeleridir. Titreşimli yüklemede alt gerilme sıfırdır, tam değışken yüklemede ise ortalama gerilme sıfırdır.

$$\sigma_{ortalama}(\text{mean}) = \frac{\sigma_{üst} + \sigma_{alt}}{2}$$

$$\sigma_{genlik}(\text{alternating}) = \frac{\sigma_{üst} - \sigma_{alt}}{2}$$

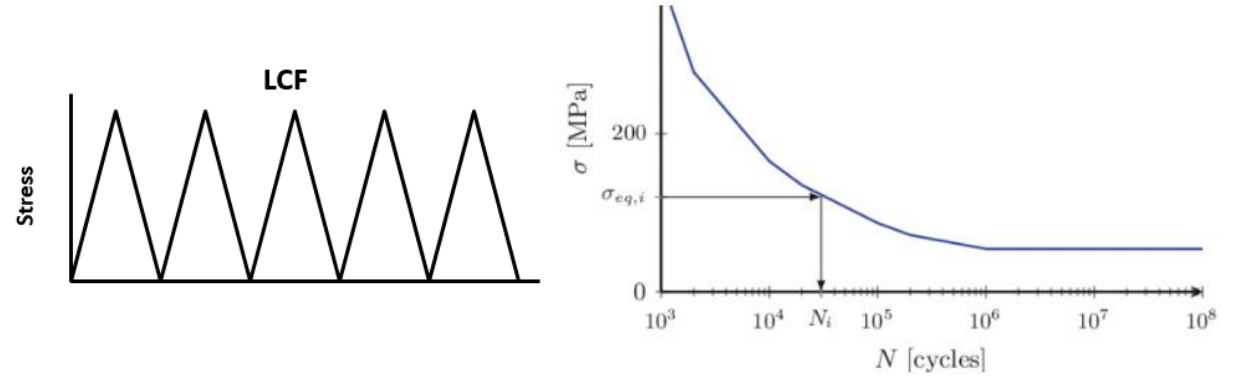
Life - Fatigue

Wöhler Eğrisi

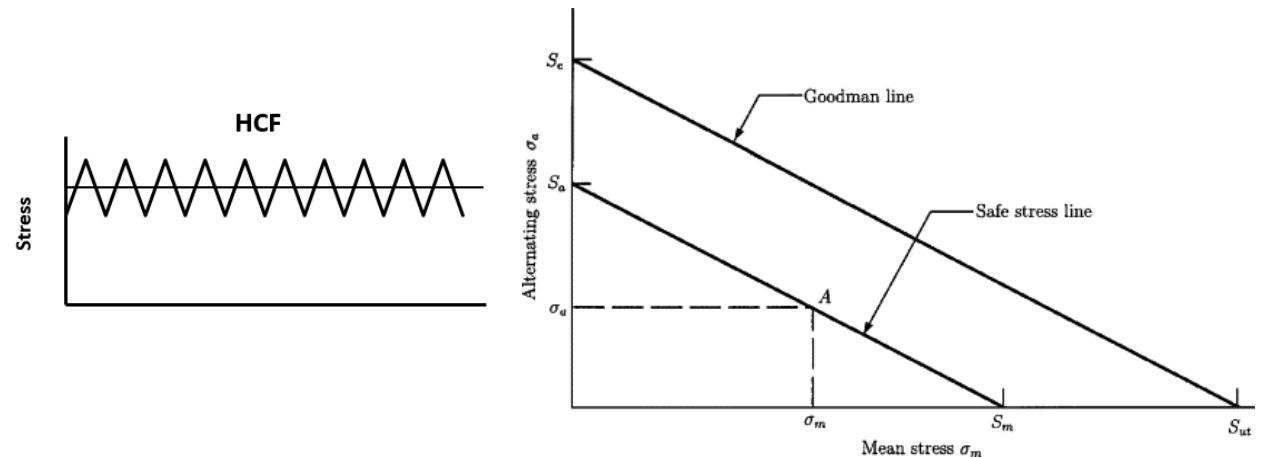


Wöhler eğrisi (S-N Curve), gerilme genliği ile yük tekrarı arasındaki bağıntıyı veren eğridir. Belirli bir yük tekrarından sonra malzemenin kırılmasına neden olan gerilmeleri içeren bölgeye **zaman mukavemet bölgesi**, kırılmanın görülmeyen bölgeye **sürekli mukavemet bölgesi** adı verilir. Her wöhler eğrisi sabir bir ortalama gerilmeye göre oluşabilecek sonuçları vermektedir. Bu nedenle malzemelerin bütün dinamik durumlar için deneylerinin yapılması ve ayrı ayrı Wöhler eğrilerinin çizilmesi gerekir.

Düşük Çevrimli Yorulma (Low Cycle Fatigue)



Yüksek Çevrimli Yorulma (High Cycle Fatigue)

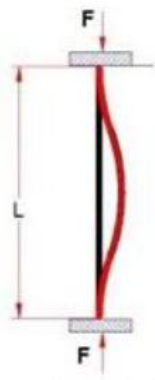


Buckling



Her iki uç mafsallı
K=1

$$F_{cr} = \frac{\pi^2 * E * I}{L^2}$$



Her iki uç sabit
K=0.25

$$F_{cr} = \frac{\pi^2 * E * I}{0.25 * L^2}$$



Bir uç mafsallı
Bir uç sabit
K=0.5

$$F_{cr} = \frac{\pi^2 * E * I}{0.5 * L^2}$$



Bir uç serbest
Bir uç sabit
K=4

$$F_{cr} = \frac{\pi^2 * E * I}{4 * L^2}$$

Hesaplamak için 3 metod vardır:

1. Bruhn Teori
2. Eigenbuckling Analiz
3. Nonlinear Buckling Analiz

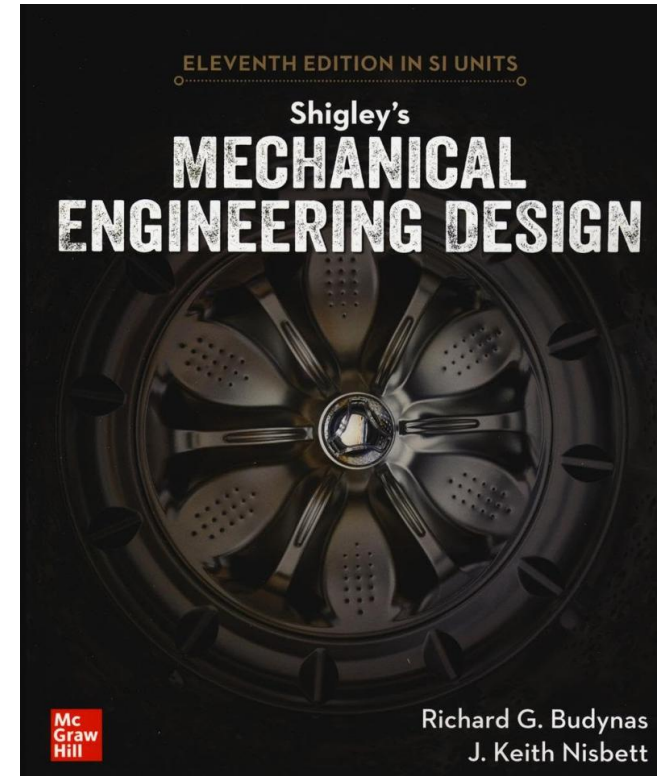
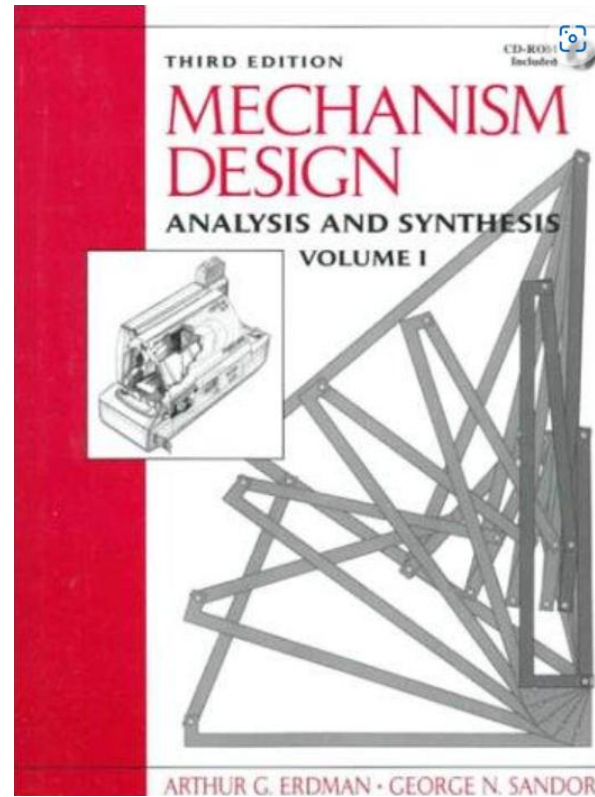
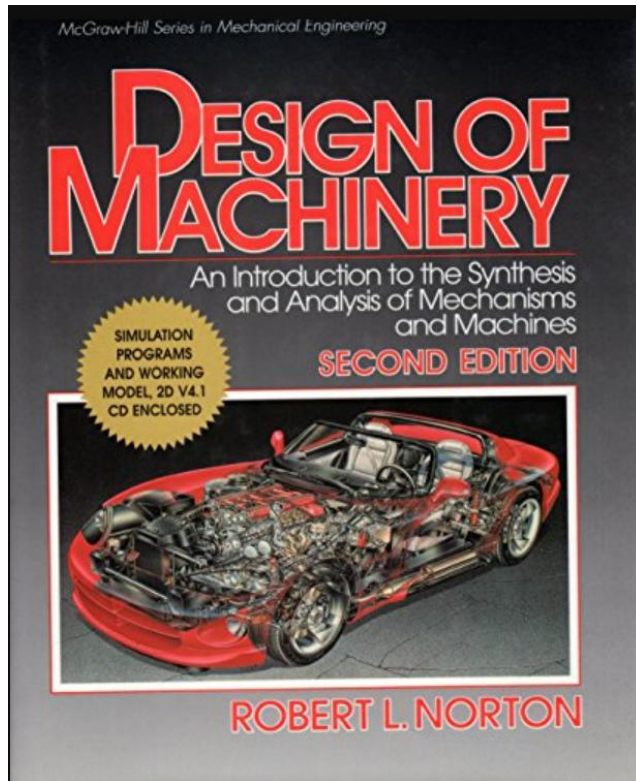
Bu hesaplamalarda "Slenderness ratio" bakılmalı, 120'den düşük ise 1 ve 2. metod için emniyet katsayısı 5 alınmalıdır. 120'den büyük ise emniyet katsayısına gerek yoktur. 3. Metod için emniyet katsayısı hiçbir durumda yoktur.

- İnce cidarlı parçalar, basma kuvvetlerine ya da yüksek basınç yüklerine maruz kalan parçalar içi kritiktir.
- Mekanik veya yapısal bir eleman eksenel olarak basma kuvveti altında iken, kuvvet belli bir kritik büyüklüğe ulaşınca eleman ani bir deformasyona uğrar. Bu fenomen burkulma (Buckling) olarak adlandırılır.
- Kritik yük büyüklüğü (F_{cr});
 - Mekanik elemanın yapıldığı malzemeye,
 - Geometrik ölçülerine,
 - Mafsal durumuna göre değişir.

Structural Analyses Expectations

- All assumptions and simplifications should be explained clearly.
- Only static analysis is expected. No dynamic analysis is required.
- No non-linear analysis is expected.
- Parts can be analyzed as isolated. A system level global analysis is not expected. But, an assessment is expected which explains “what would happen if all parts were analyzed as a system”.
- For simple parts (e.g. 2 force members), hand calculations are seen enough.
- All boundary conditions, loadings, contacts should be explained clearly.

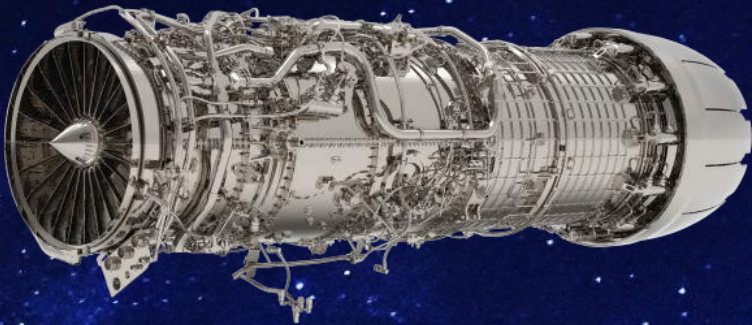
Sources



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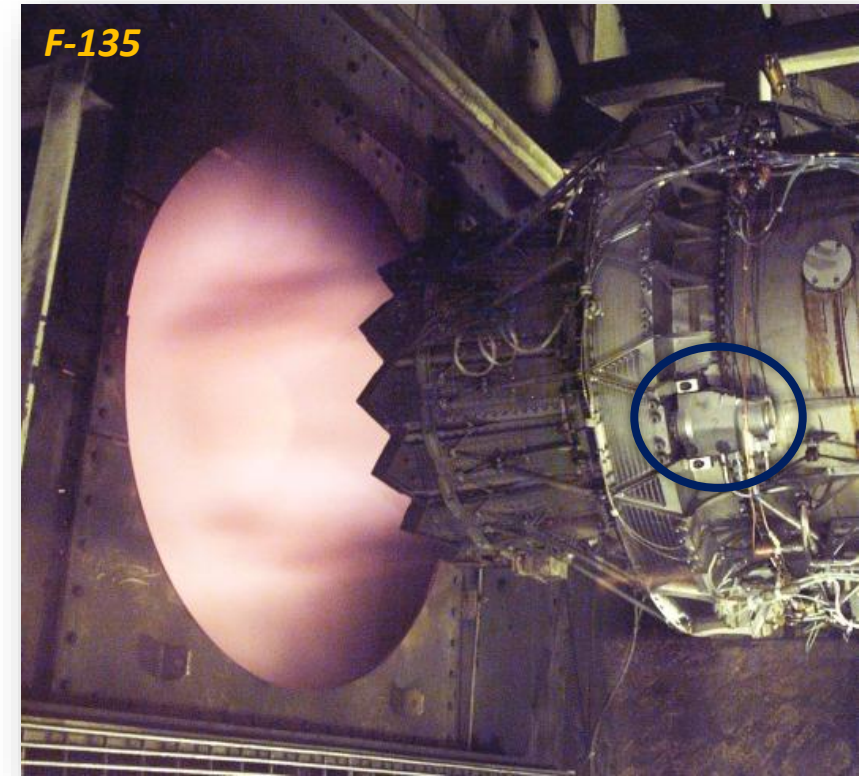
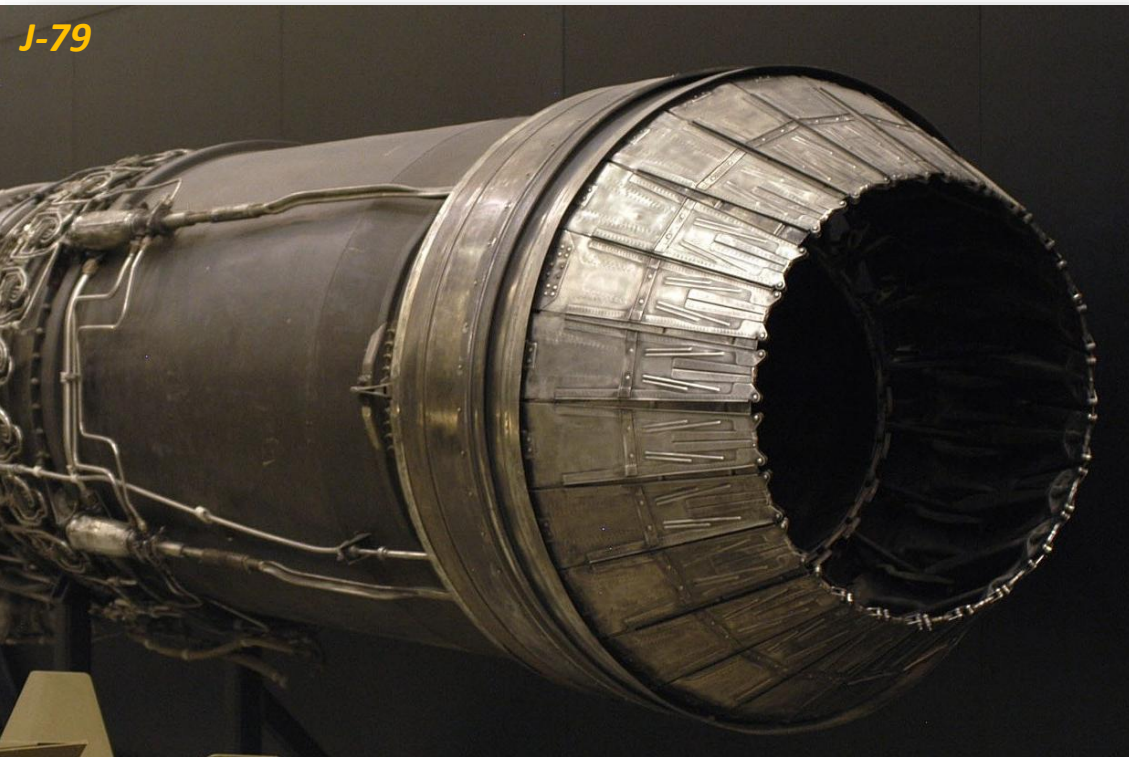
Brief Introduction of Nozzle Actuation Systems

Cengiz Öztürk



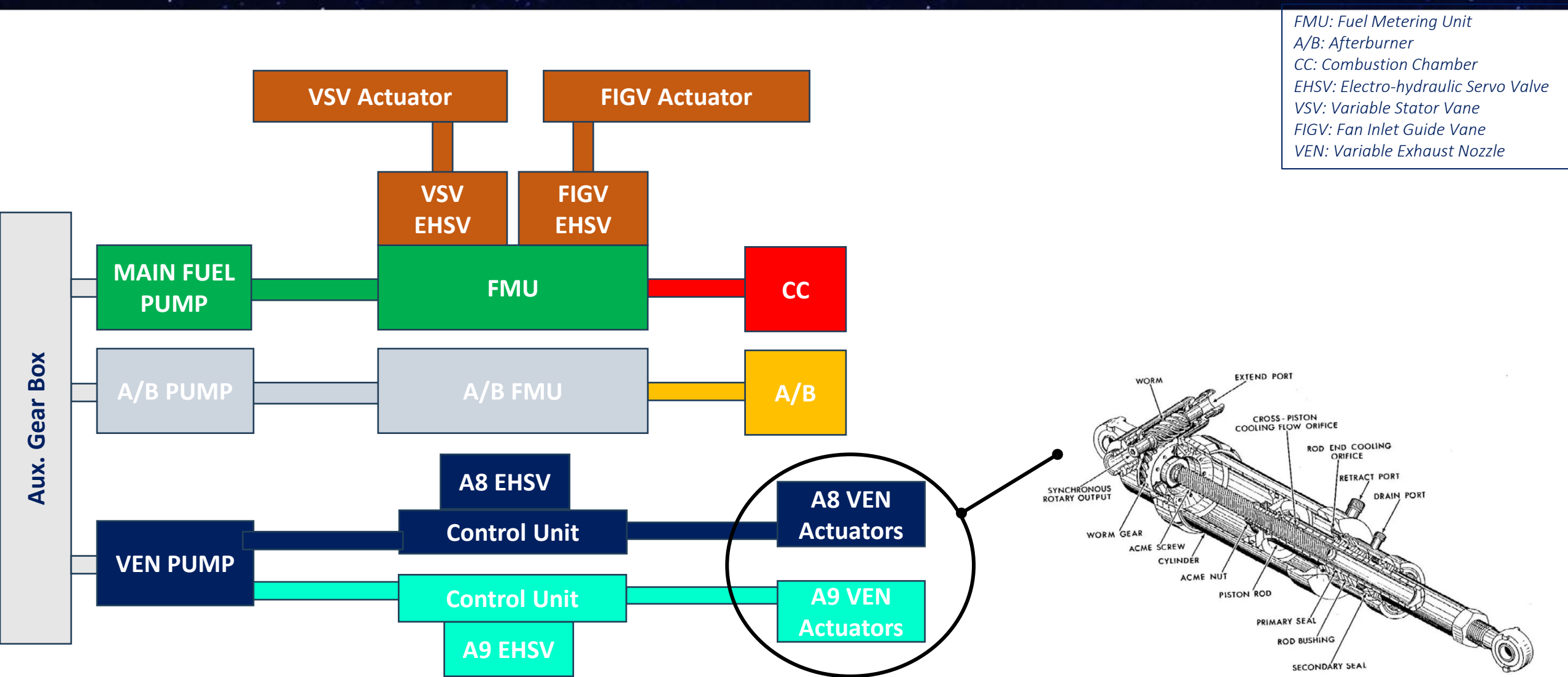
What drives variable geometries

- In turbofan engines, variable geometries (Fan Inlet guide vanes, Variable stator vanes, variable exhaust nozzle) are driven by **hydraulic or fuel**.



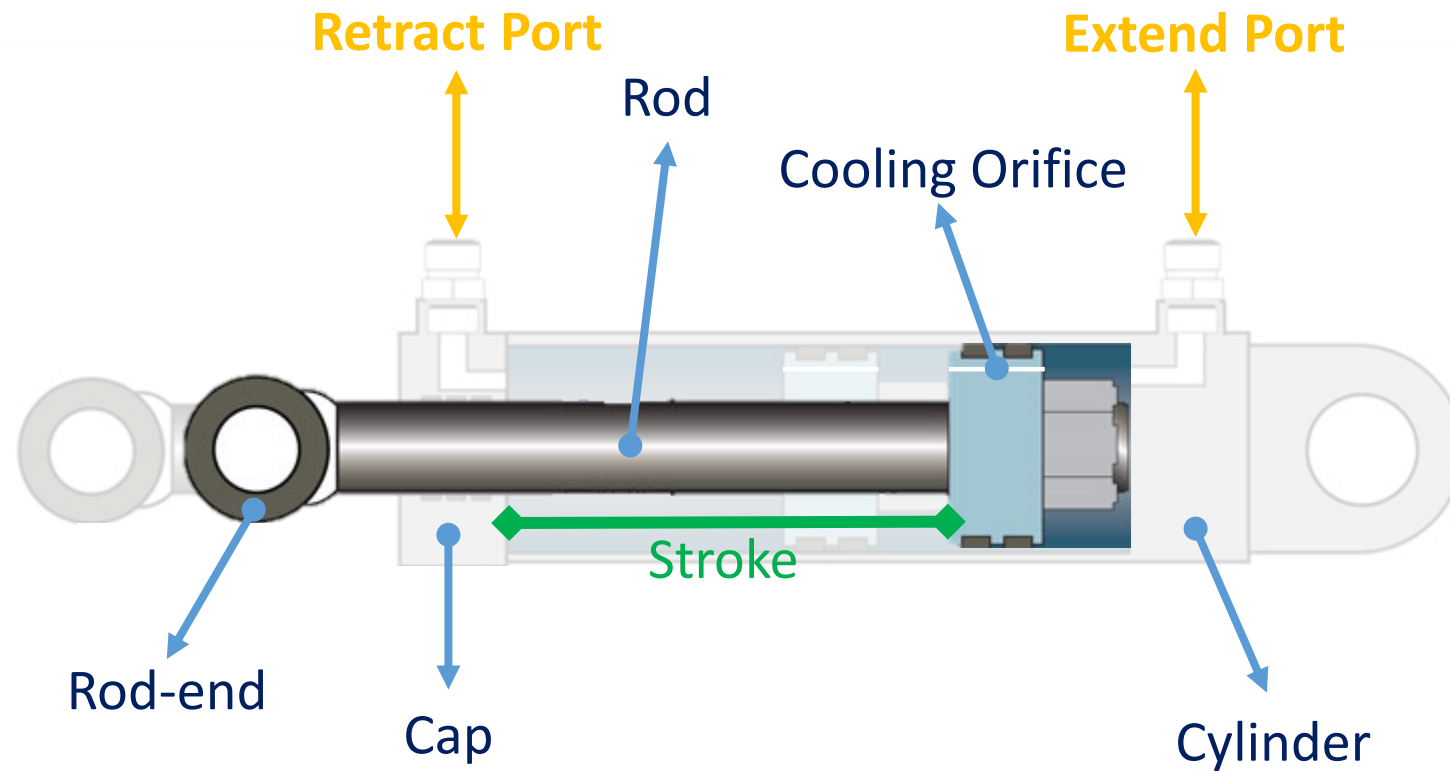
These images were taken from Google.

Basic Fuel System of a Engine

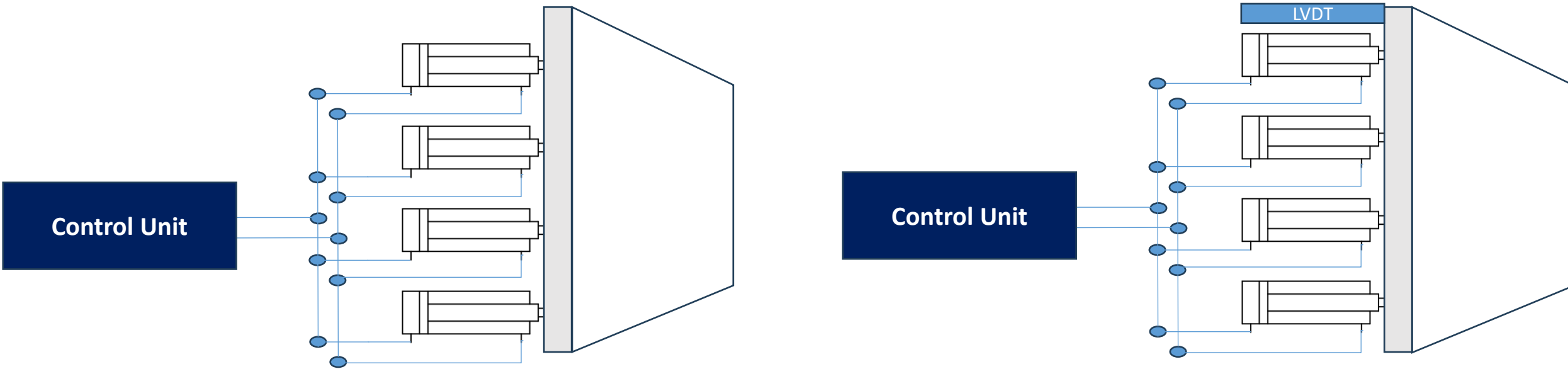


https://www.parker.com/content/dam/Parker-com/Online/Product-Images/Hydraulic-Systems-Division/zoom_1000x1000/actuatorSystem2_1000W_alt1zm.jpg

Hydraulic Cylinder / Actuator

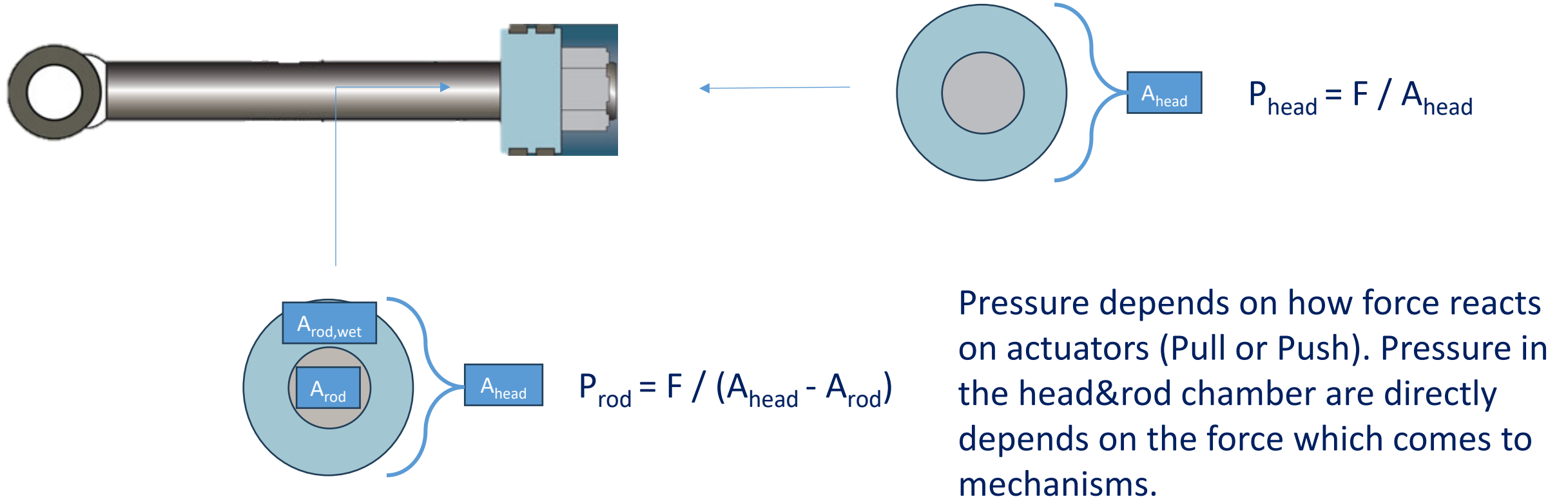


Nozzle Mechanisms with Actuators

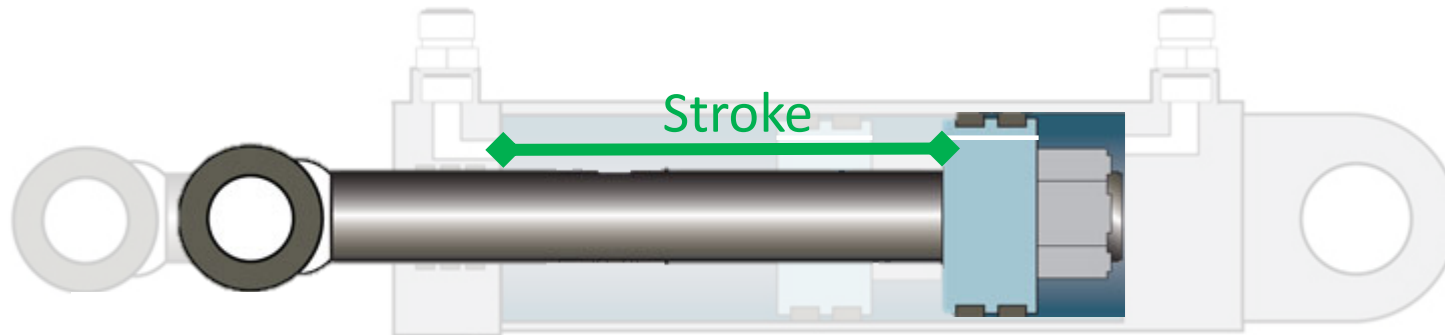


- Why actuators hydraulic connections are connected same tube for all actuator ?
- How does system know and control that what is the A8 area and where it should be stayed during engine in operation ?

Basic Actuator Calculation



Basic Flow Rate Calculation



Head and Rod Volumes;

$$V_{\text{head}} = \text{Stroke} \times A_{\text{head}}$$

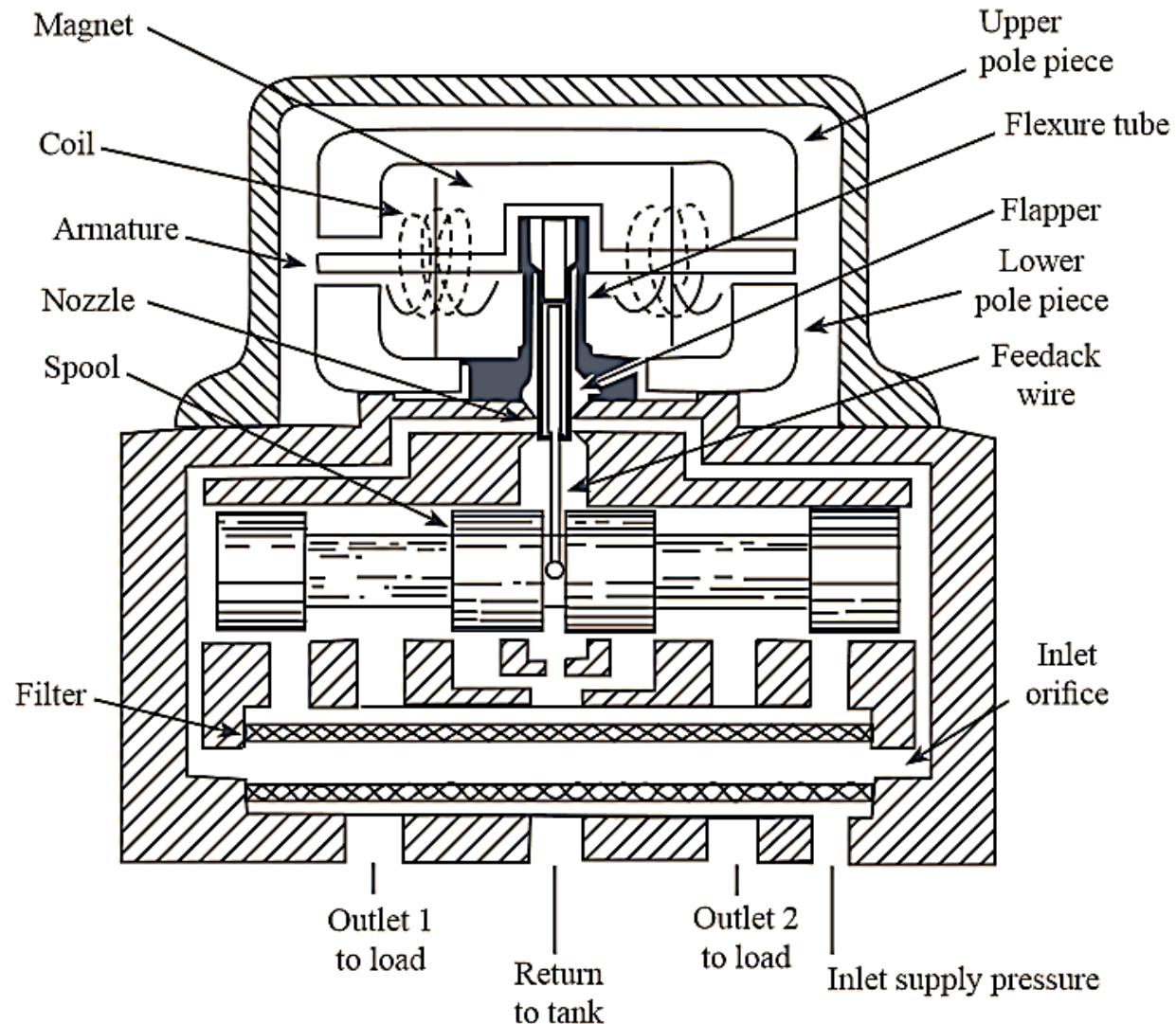
$$V_{\text{rod}} = \text{Stroke} \times A_{\text{rod}}$$

Required Flow Rate from Pump

$$Q_{\text{head}} = V_{\text{head}} / t_{\text{requirement}}$$

$$Q_{\text{rod}} = V_{\text{rod}} / t_{\text{requirement}}$$

What is electro-hydraulic servo valve



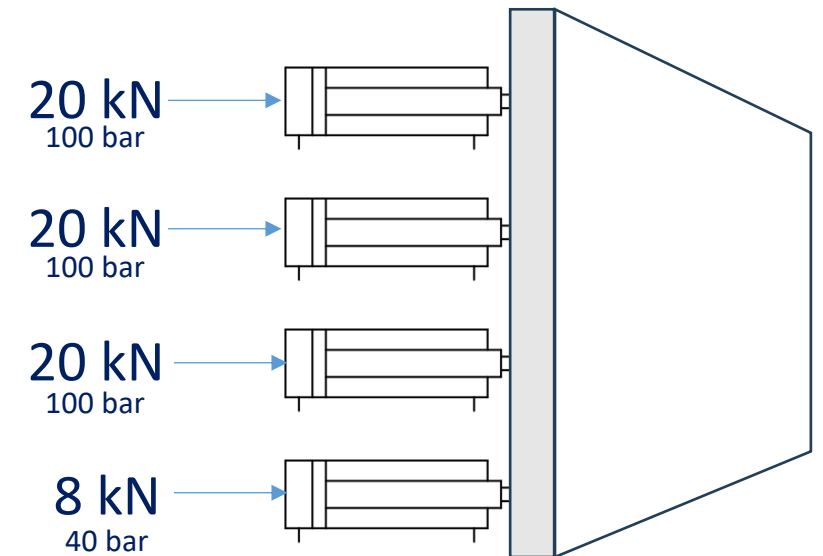
- When current is applied torque motor is rotated. According to its rotation pressure increase/decrease in the spools behind chambers.
- According to this, spool shifts to left or right. This causes the feedback wire to bend and creates a force to hold the spool at a same place. This leads to flow rate to one of the outlets. At the same time the other outlet port is also connected to the return port. That movement creates motion in actuators.

Synchronization



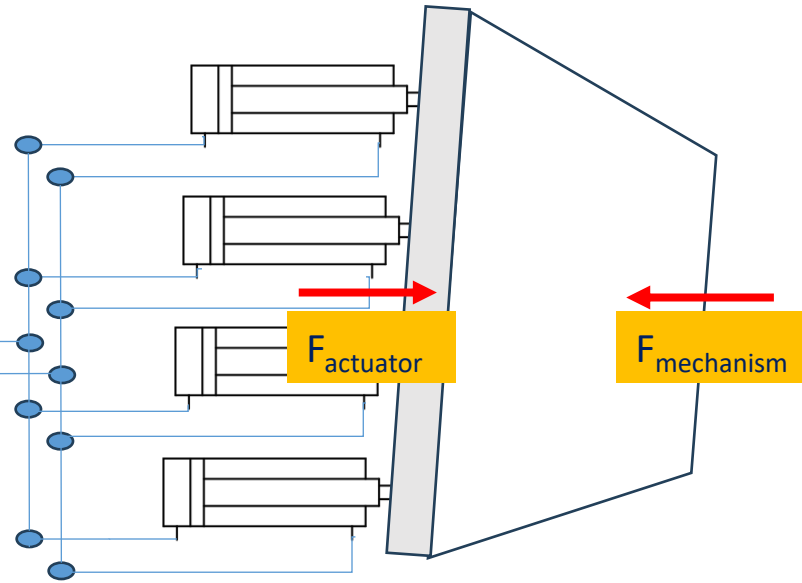
How is synchronization achieved?

- All actuators pressure are same only **theoretically**.
- But there are frictions, thermals, tolerances.
- What would happen if one actuator internal leakages are increased so much ?



The image was taken from Google.

Synchronization



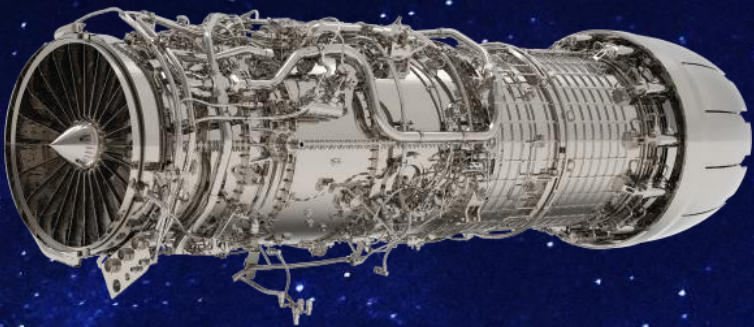
- What would happen during maneuver ?

- All Actuator push and pull fittings are connected to same feeding tube. So **pressure and forces comes from actuator must be the similar.**
- Mechanism stiffness must be cover that differences which caused by force variations, internal leakages at a some point in actuator etc. Mechanisms components are bended and deflected and thanks to that actuators could be held at a small stroke differences. These bending and deflections causes additional forces for the actuator which is stayed back because of forces differences or pressure changes in actuators(increased internal leakage). Otherwise system can't be controlled.

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Brief Introduction to Nozzle Performance

Furkan Göktürk Öztiryaki

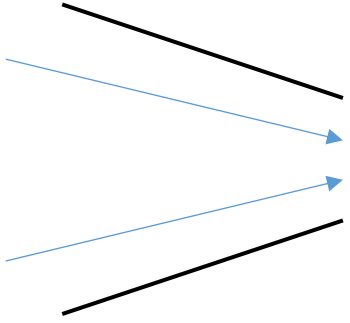


What is a Nozzle?

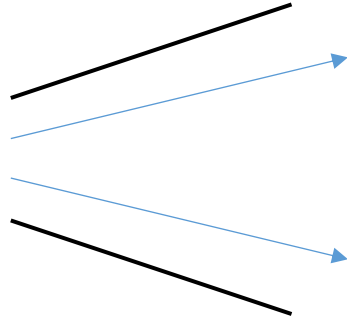
- Used in controlling the fluid flow
- Changes the flow speed to desired values
- Changes the flow pattern



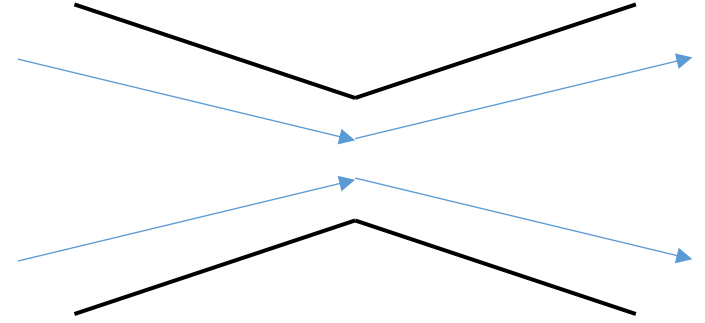
Nozzle Types



Convergent Nozzle



Divergent Nozzle



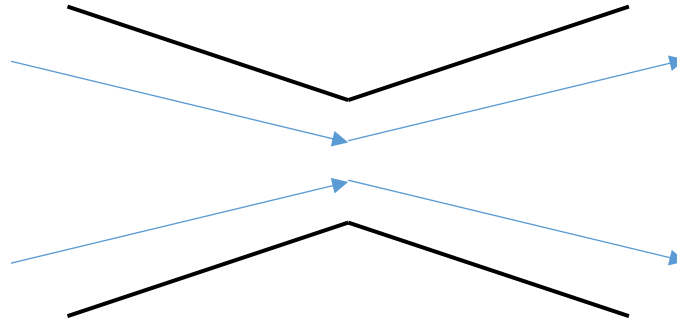
Convergent-Divergent Nozzle

Con-Di Nozzle

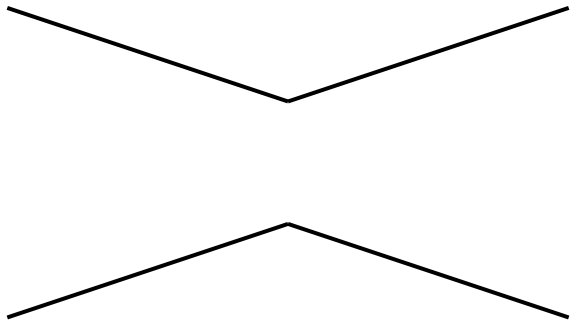
CD Nozzle

Supersonic Nozzle

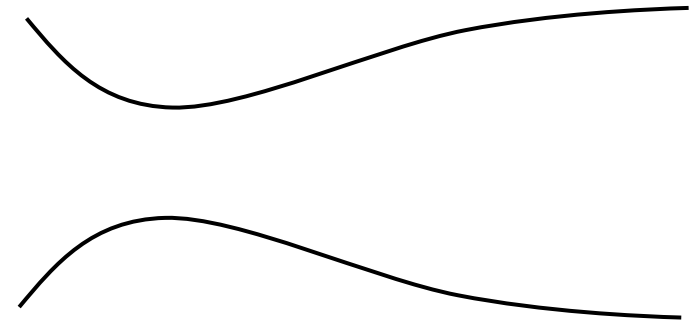
Nozzle Types



Convergent-Divergent Nozzle

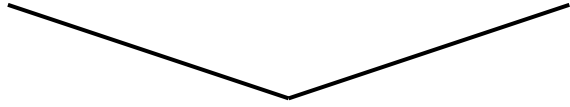


Aircraft nozzle
Sharp contours
Variable geometry

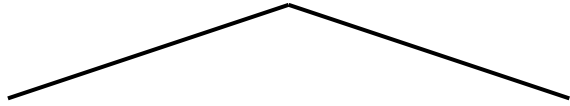


Rocket nozzle
Smooth contours
Constant geometry

Nozzle Types



Aircraft nozzle
Sharp contours
Variable geometry

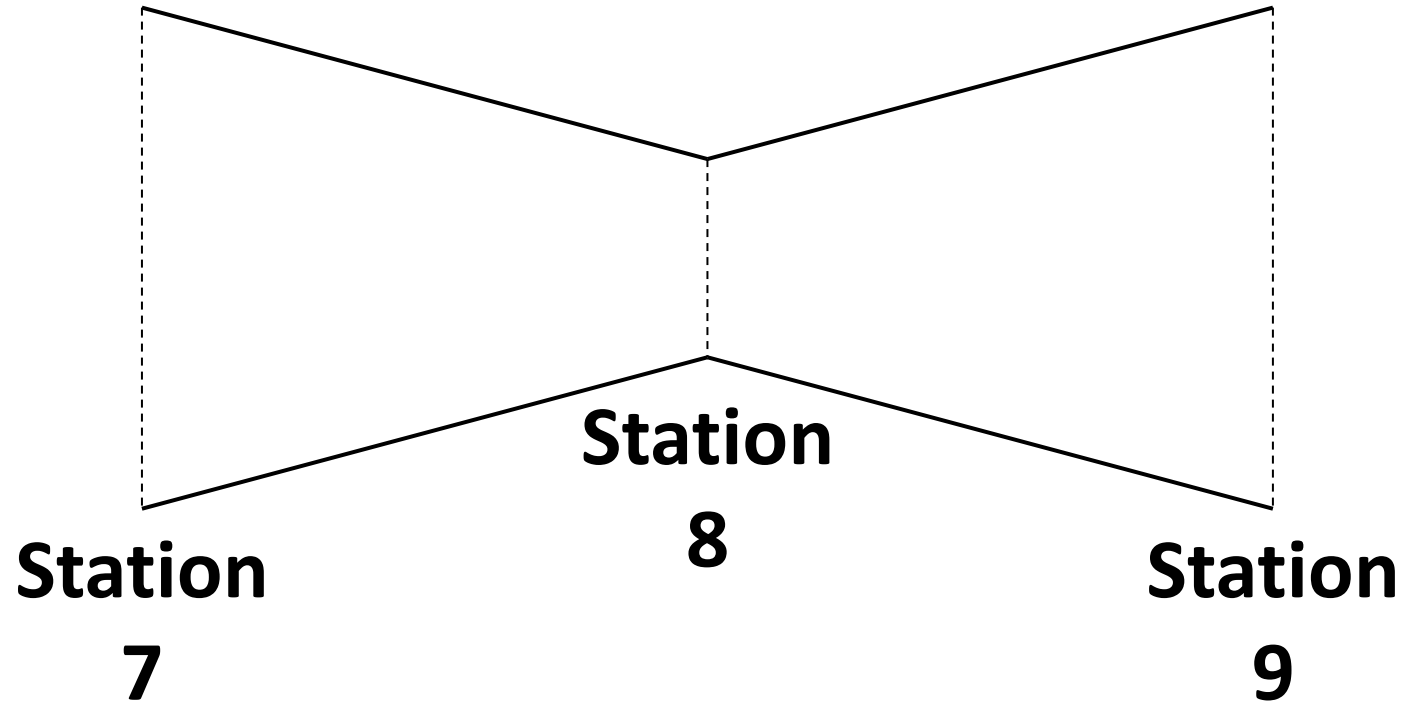


Rocket nozzle
Smooth contours
Constant geometry

- Which one is more efficient?
- Why both do not use the same?

Stations

- Station 7: Nozzle inlet
- Station 8: Nozzle throat
- Station 9: Nozzle exit
- SAE-AS755



Corrected Parameters

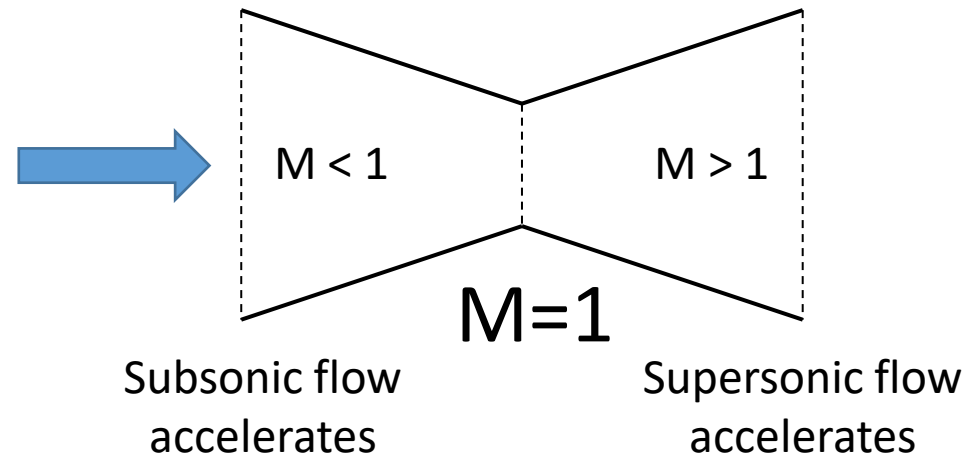
- Corrected parameters are used to describe the flow in a simple manner
- Let's start with the definition of Mach number
- $M = \frac{V}{\sqrt{\gamma RT_s}}$
- Conservation of mass: $\dot{m} = \rho VA$
- $M = \frac{\dot{m}}{\rho A \sqrt{\gamma RT_s}}$
- Ideal Gas Law (Equation of State): $P = \rho RT$
- $M = \frac{\dot{m}}{\frac{P_s}{RT_s} A \sqrt{\gamma RT_s}} = \frac{\dot{m} \sqrt{T_s}}{P_s} \sqrt{\frac{R}{\gamma} \frac{1}{A}}$

Corrected Parameters

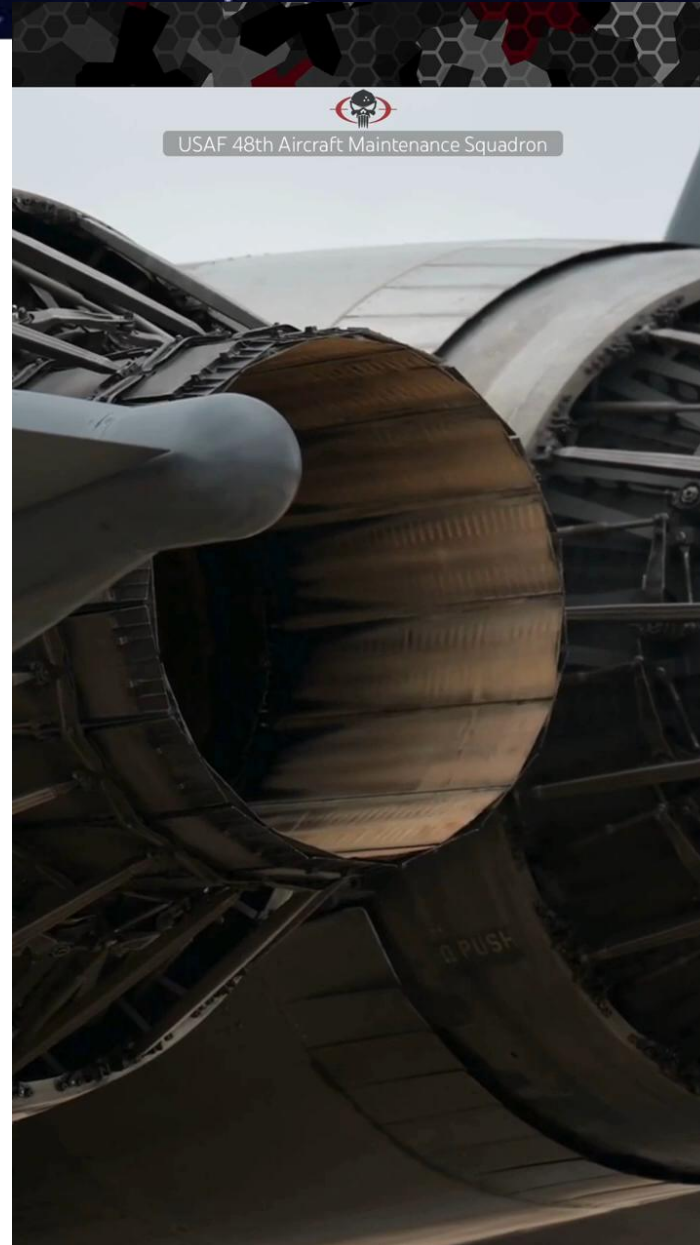
- $$M = \frac{\dot{m}}{\frac{P_s}{RT_s} A \sqrt{\gamma RT_s}} = \frac{\dot{m} \sqrt{T_s}}{P_s} \sqrt{\frac{R}{\gamma}} \frac{1}{A}$$
- Isentropic flow relations: $\frac{P_s}{P_t} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{-\frac{\gamma}{\gamma-1}}, \frac{T_s}{T_t} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{-1}$
- $$M = \frac{\dot{m} \sqrt{T_t}}{P_t} \frac{1}{A} \sqrt{\frac{R}{\gamma}} \left(1 + \frac{\gamma-1}{2} M^2\right)^{-\frac{1}{2}} \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma}{\gamma-1}}$$
- $$M = \frac{\dot{m} \sqrt{T_t}}{P_t} \frac{1}{A} \sqrt{\frac{R}{\gamma}} \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma+1}{2(\gamma-1)}}$$
- Corrected flow is defined as $W_c = \frac{\dot{m} \sqrt{T_t/T_{t,ref}}}{P_t/P_{t,ref}}$

Mass Flow Choking

- $W_c = \frac{\dot{m} \sqrt{T_t/T_{t,ref}}}{P_t/P_{t,ref}}, W_p = \frac{\dot{m} \sqrt{T_t}}{P_t}$
- Corrected flow is an approximation on Mach number
- Let's assume that Mach number is 1
- ~~$M = W_p \frac{1}{A} \sqrt{\frac{R}{\gamma}} \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma+1}{2(\gamma-1)}}$~~ **Constants**
- Only way to change the corrected flow when the Mach number reaches 1 is by changing the area, this is called **choking**

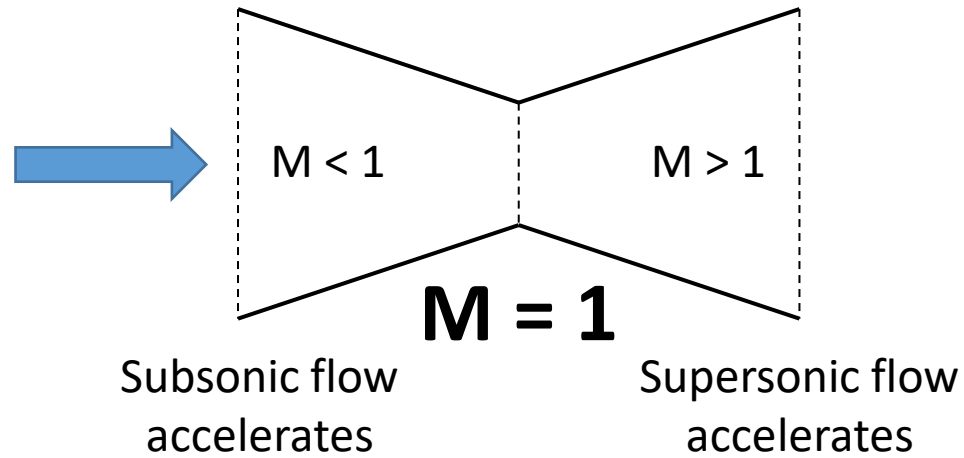


Mass Flow Choking



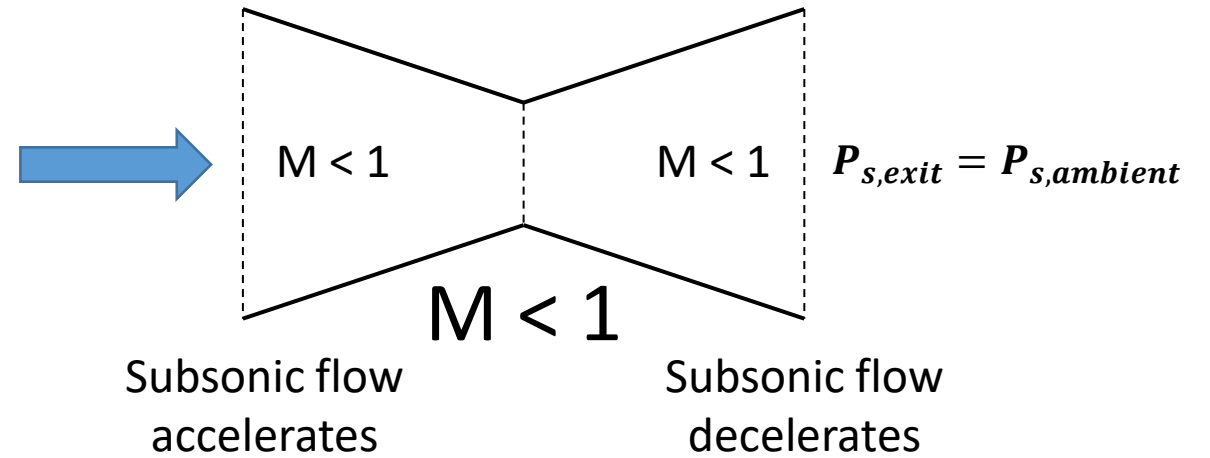
Why the
nozzle area
changes when
AB is lit?

Choked & Unchoked Nozzle



Choked Nozzle

- Converging part is subsonic
- The narrowest area reaches sonic
- Diverging part is supersonic



Unchoked Nozzle

- Converging part is subsonic
- The narrowest area does not reach sonic
- Diverging part is also subsonic and behaves as

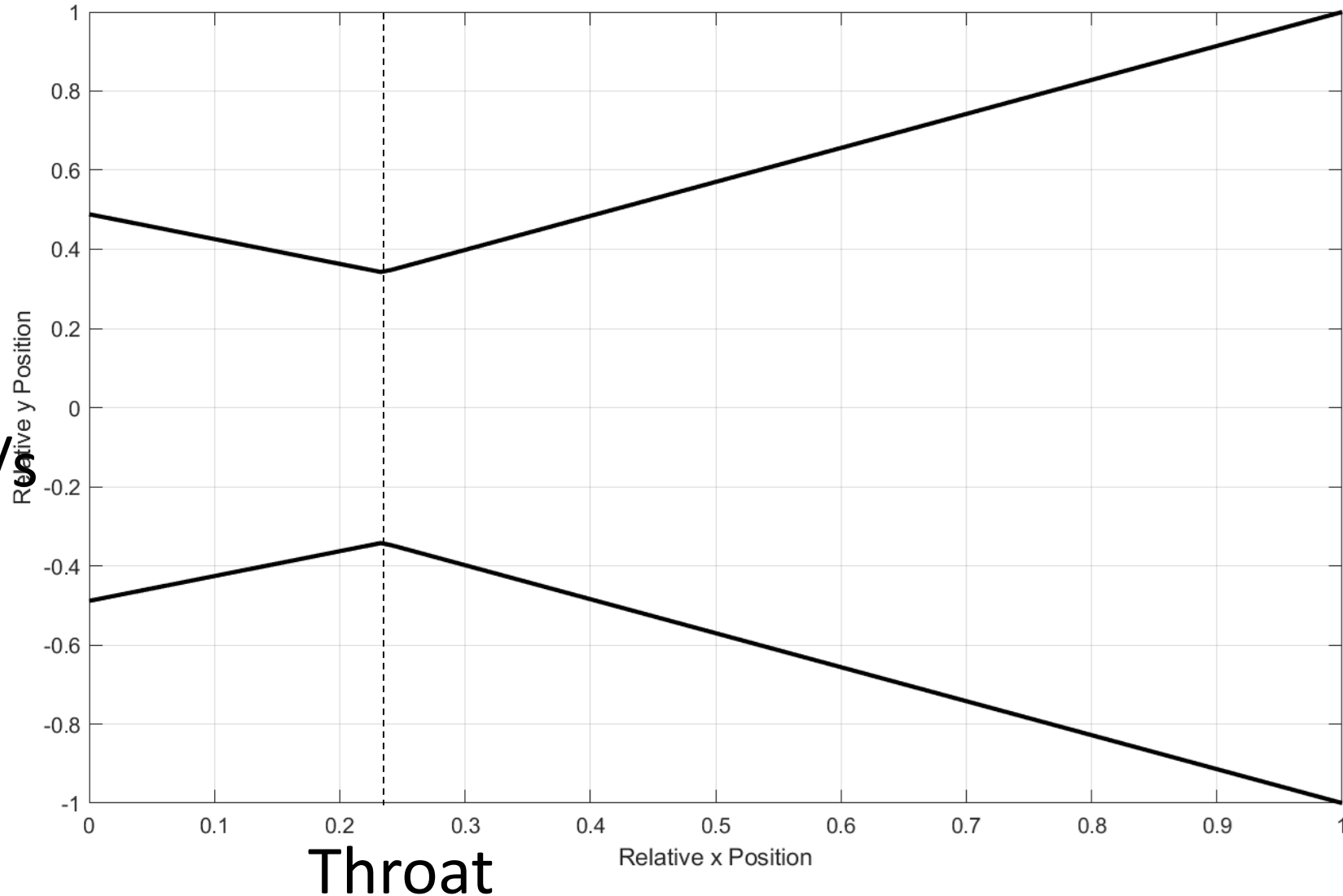
Mass Flow Choking

- How do we check for choking?
- $\frac{P_s}{P_t} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{-\frac{\gamma}{\gamma-1}}$
- Lets assume $\gamma = 1.33$, and $M = 1$
- $\frac{P_t}{P_s} = 1.85$
- How do we apply this to the nozzle?
 - If: $\frac{P_t}{P_{s,ambient}} > 1.85 \rightarrow$ choked nozzle
 - Else: $\frac{P_t}{P_{s,ambient}} < 1.85 \rightarrow$ unchoked nozzle
- $\frac{P_t}{P_{s,ambient}}$ is called Nozzle Pressure Ratio (NPR)

Nozzle Expansion

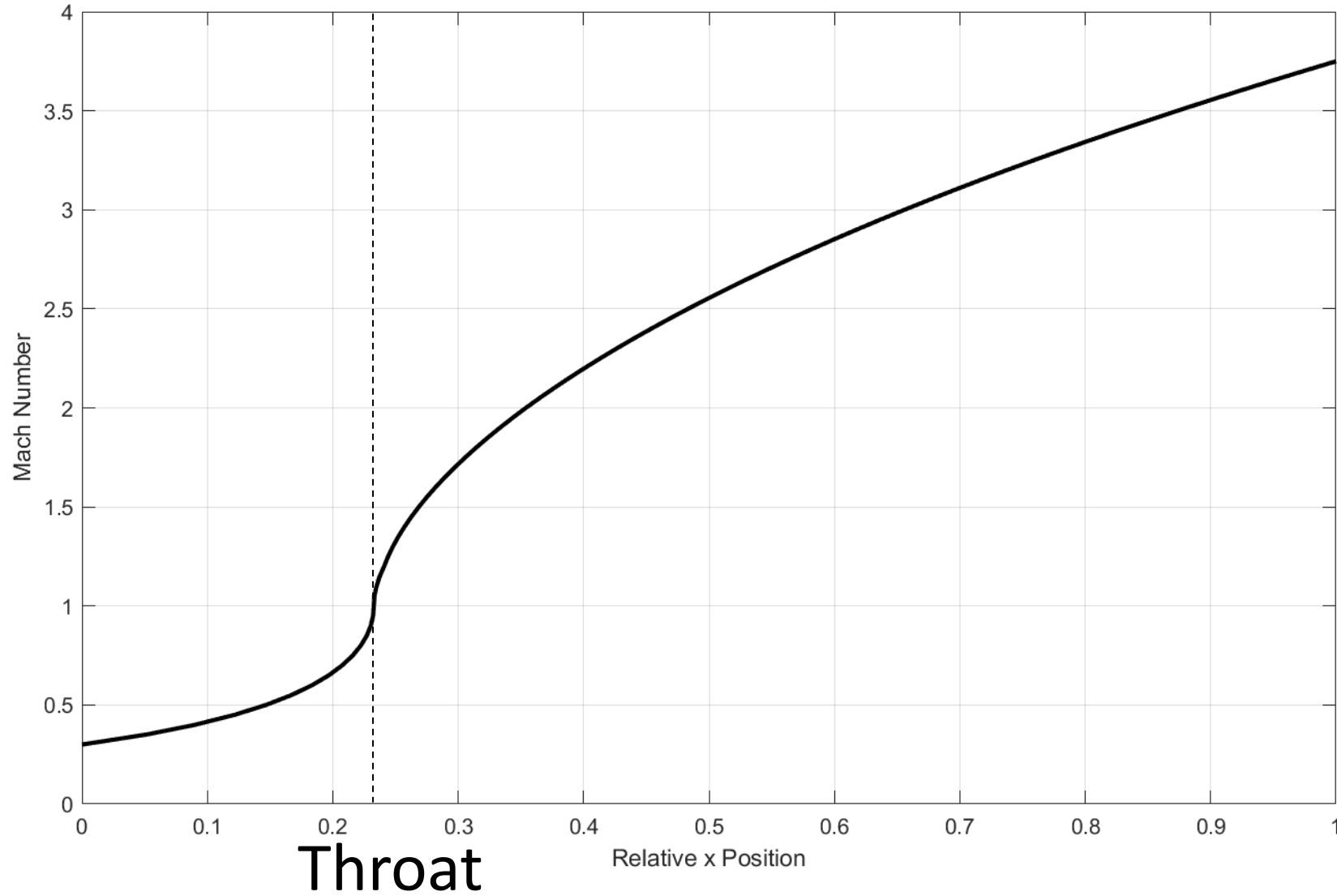
- Let's consider a hypothetical nozzle geometry

NPR = 20
 $\dot{W}_7 = 100 \text{ kg/s}$
 $T_{t7} = 288 \text{ K}$



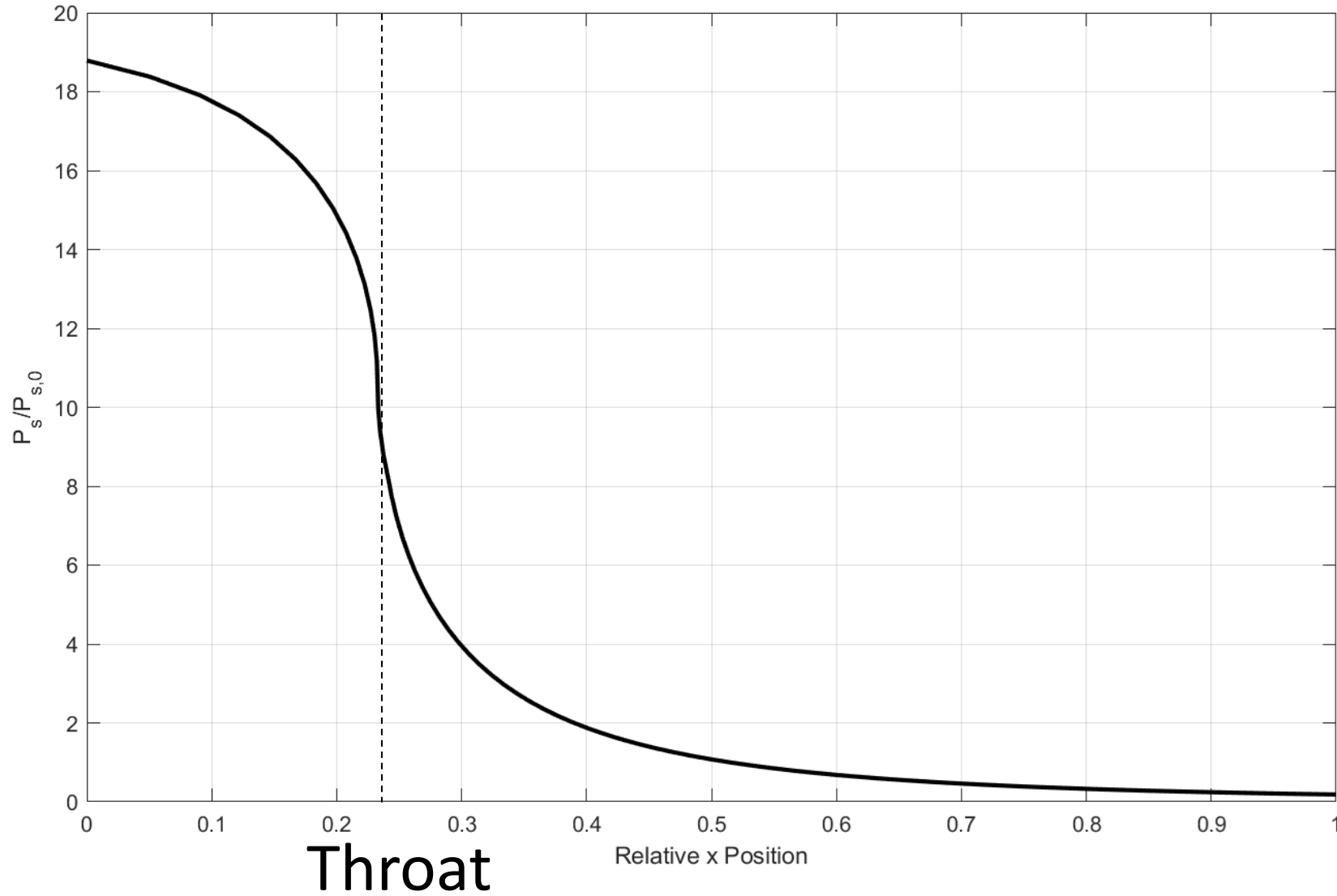
Nozzle Expansion

- Calculate the Mach number using the isentropic flow relations



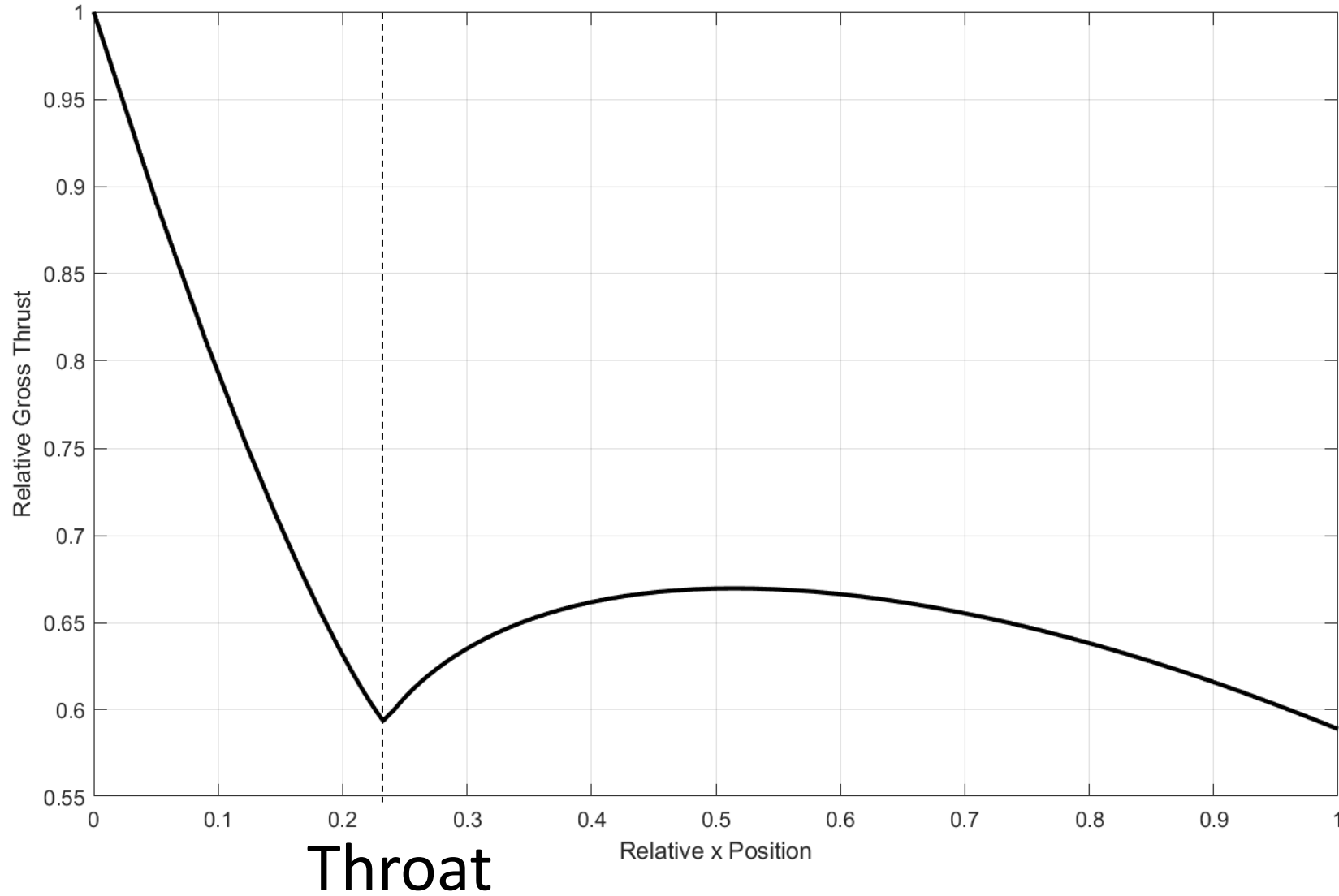
Nozzle Expansion

- Calculating the static pressure at each position relative to the ambient



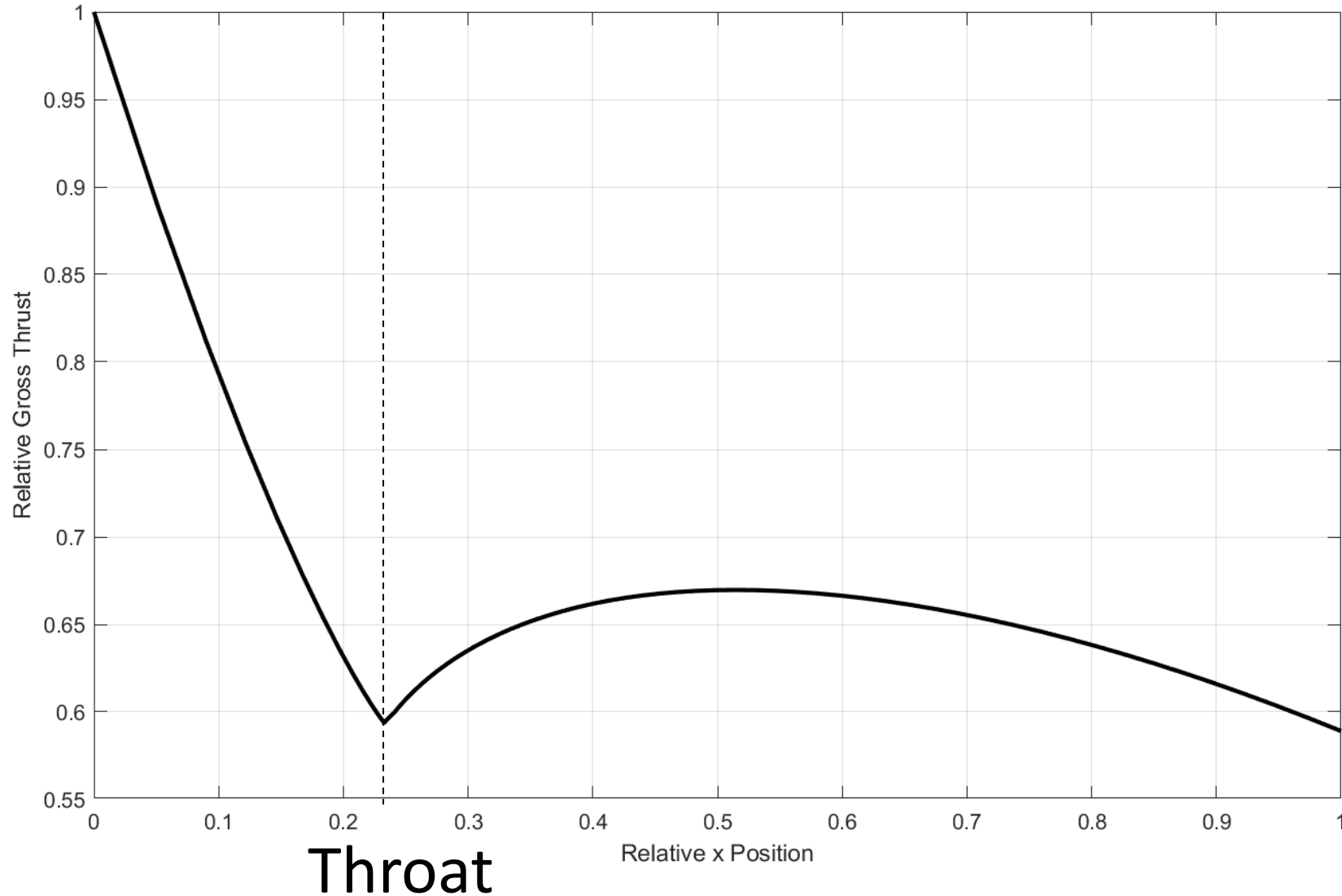
Nozzle Expansion

- Calculate the thrust at each position



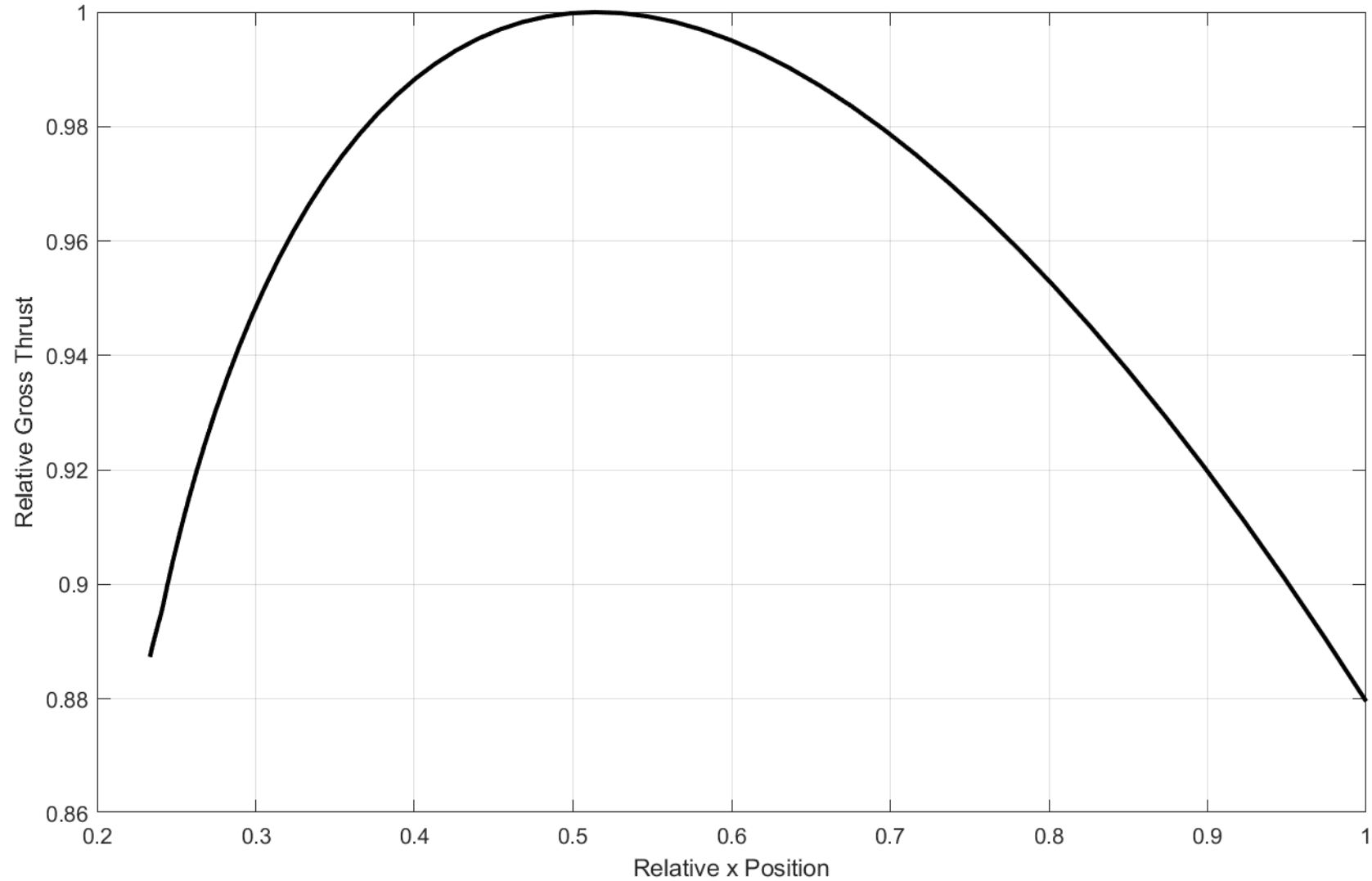
Nozzle Expansion

- Highest thrust is at the inlet of the nozzle, so **why not just use a constant area nozzle?**



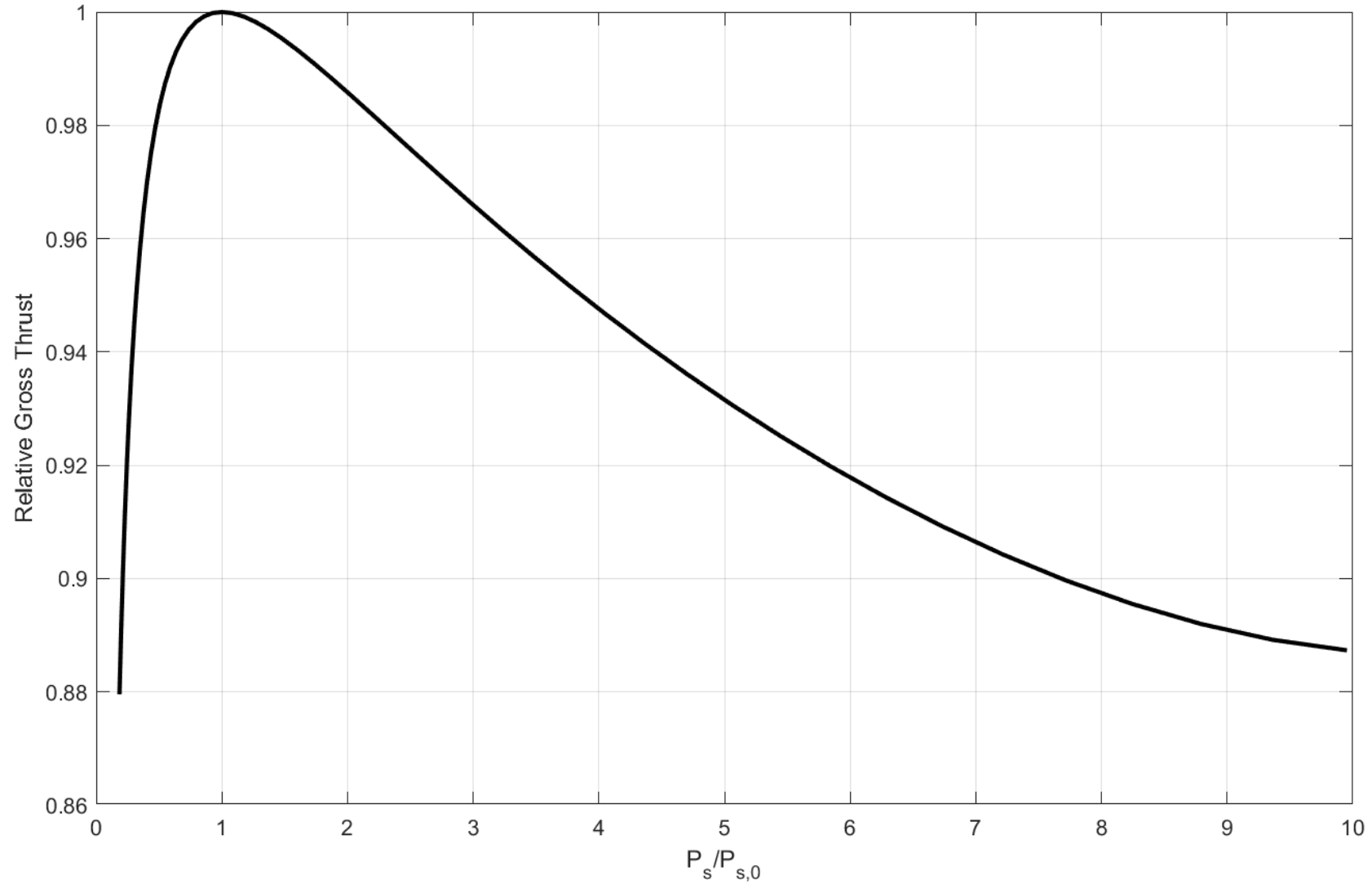
Nozzle Expansion

- The gross thrust seems to make a peak

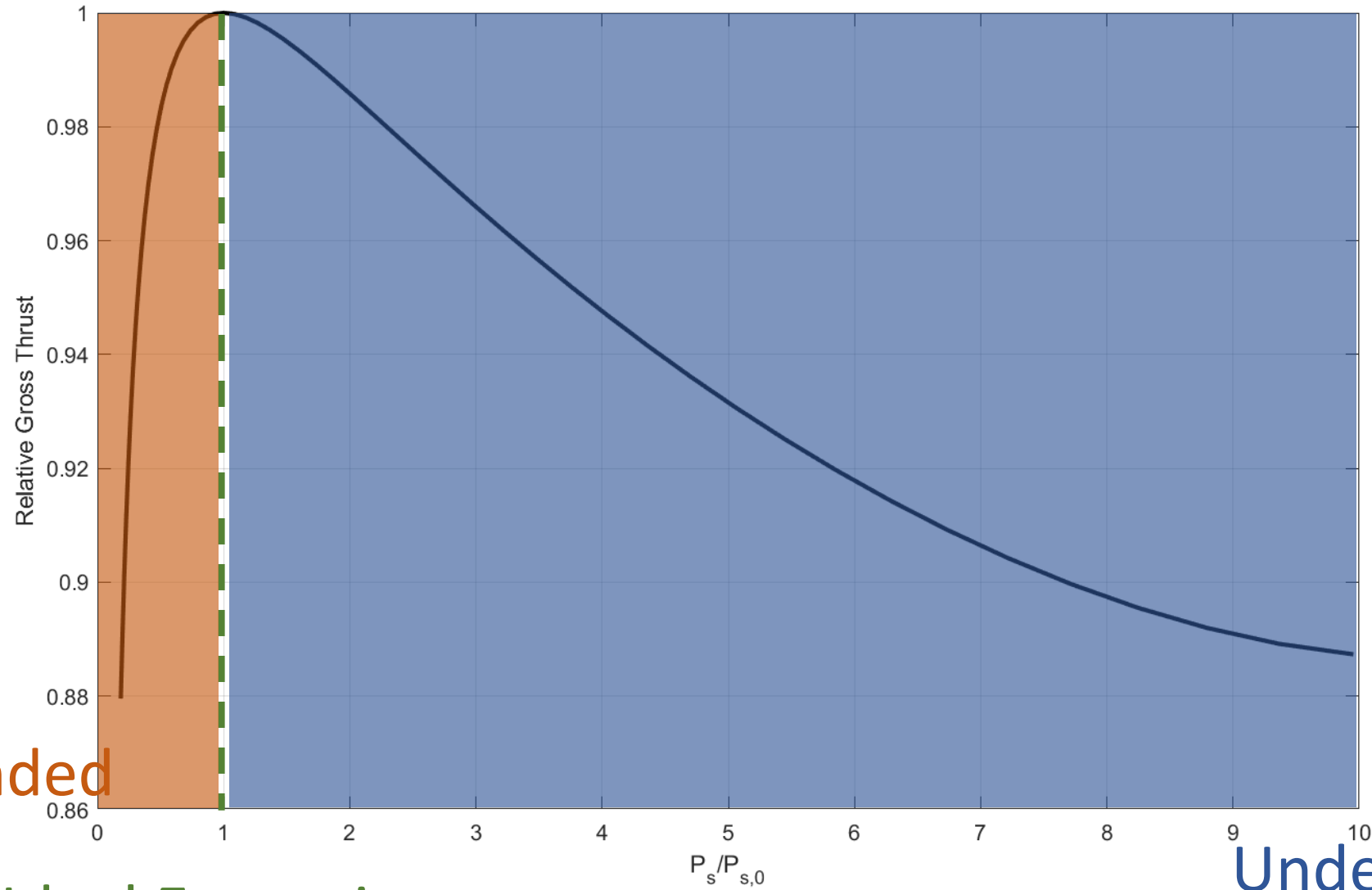


Nozzle Expansion

- Changing the x axis to $P_s/P_{s,0}$, the highest thrust is obtained when the $P_s = P_{s,0}$



Nozzle Expansion



Over Expanded

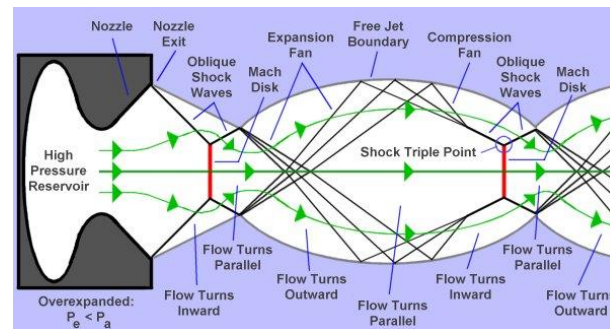
Ideal Expansion

Under Expanded

Nozzle Expansion



Nozzle Expansion



Calculation of Thrust and Cfg

- Gross thrust is defined as:
- $F_g = \dot{m}V_{9,ax} + A_9(P_{s,9} - P_{s,0})$
- Gross Thrust Coefficient (C_{fg}) is defined as:
- $C_{F_g} = \frac{F_g}{F_{g,ideal}}$
- The actual gross thrust is obtained from tests & analyses, while the ideal gross thrust is calculated
- In order to calculate the ideal gross thrust, the nozzle is assumed to be ideally expanded, without any losses (using the total properties at the inlet)

Calculation of C_d

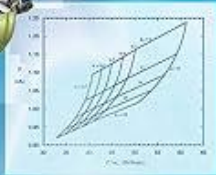
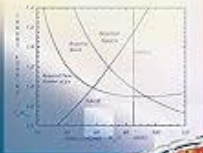
- The ideal mass flow rate that can pass through the throat area differs from the actual one
- A coefficient, called discharge coefficient (C_d) is defined to correct this
- C_d is defined as the ratio of effective throat area to the actual throat area
- $$C_d = \frac{A_{g,e}}{A_g}$$
- The effective throat area is the area that chokes the flow in an ideal case, while the actual area is the one that is tested or analysed

Sources

Aircraft Engine Design

Third Edition

Jack D. Mattingly
William H. Heiser
Keith M. Boyer
Brenda A. Haven
David T. Pratt



Joseph A. Schetz
Editor-in-Chief



Gas Turbine Performance

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Thank You

